

A quantitative replica-symmetric bound of Sherrington–Kirkpatrick model in the entire de Almeida–Thouless region

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Abstract

We consider the Sherrington–Kirkpatrick model with inverse temperature $\beta > 0$ and deterministic external field $h > 0$. Let q be the replica-symmetric fixed point defined by $q = \mathbb{E} \tanh^2(h + \beta\sqrt{q} Z)$, where Z is a standard Gaussian random variable. We prove that, uniformly on compact subsets of the strict de Almeida–Thouless region, defined by $\beta^2 \mathbb{E} \operatorname{sech}^4(h + \beta\sqrt{q} Z) < 1$, the overlap satisfies the concentration estimate:

$$\mathbb{E} \langle (R_{12} - q)^2 \rangle = O(N^{-1}).$$

As a consequence, we obtain an $O(N^{-1})$ error bound for the replica-symmetric free-energy and identify the finite-volume replicon susceptibility.

The present paper provides an alternative proof of the replica-symmetric free energy formula in the de Almeida–Thouless region, which was recently established by Lopatto [arXiv:2604.11921]. An advantage of our approach is that it establishes an explicit quantitative bound and yields the replica-symmetric free energy formula as a consequence.

Our proof is self-contained and does not use the identification of the limiting free energy with the Parisi variational formula.

1 Introduction

1.1 Model and main results

In this paper, we consider the Sherrington–Kirkpatrick model. Let $N \in \mathbb{N}$ and $\Sigma_N := \{-1, 1\}^N$. For $\beta > 0$ and $h > 0$, let

$$H_N(\sigma) := \frac{\beta}{\sqrt{2N}} \sum_{i,j=1}^N g_{ij} \sigma_i \sigma_j + h \sum_{i=1}^N \sigma_i, \quad \sigma \in \Sigma_N$$

where $(g_{ij})_{1 \leq i,j \leq N}$ are independent standard normal random variables.

Let $q = q(\beta, h)$ be the unique solution of

$$q = \mathbb{E} \tanh^2(h + \beta\sqrt{q} Z), \quad Z \sim N(0, 1), \quad (1.1)$$

and write $q := q(\beta, h)$. The existence and uniqueness for all $\beta > 0$ and $h > 0$ are obtained by a standard argument (see, for example, [19, Proposition 1.3.8]). The continuity of $q(\beta, h)$ in (β, h) follows directly from uniqueness. Indeed, if $(\beta_n, h_n) \rightarrow (\beta, h)$, every convergent subsequence of $q(\beta_n, h_n) \in [0, 1]$ has, by the dominated convergence theorem, the limit solving the fixed-point

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equation (1.1) at (β, h) . The uniqueness identifies that limit with $q(\beta, h)$, so the entire sequence converges. Set

$$r := \mathbb{E} \tanh^4(h + \beta\sqrt{q} Z), \quad (1.2)$$

and let

$$\alpha := \beta^2(1 - 2q + r) = \beta^2 \mathbb{E} \operatorname{sech}^4(h + \beta\sqrt{q} Z). \quad (1.3)$$

We work on a fixed compact set

$$\mathcal{K} \subseteq \{(\beta, h) : \beta > 0, h > 0, \alpha(\beta, h) < 1\}. \quad (1.4)$$

Then, there exists $\delta_{\mathcal{K}} > 0$ such that

$$1 - \alpha \geq \delta_{\mathcal{K}} \quad \text{on } \mathcal{K}. \quad (1.5)$$

Since the maps q, r and α are continuous on $\mathcal{K}$, the compactness and $h > 0$ imply

$$0 < q_{\mathcal{K}} := \inf_{(\beta, h) \in \mathcal{K}} q \leq \sup_{(\beta, h) \in \mathcal{K}} q < 1. \quad (1.6)$$

This inequality follows from the facts that $q = 0$ would contradict (1.1) with $h > 0$, and that $q < 1$ follows from the inequality $\tanh^2(h + \beta\sqrt{q} Z) < 1$.

For $0 \leq s \leq 1$, define the smart-path Hamiltonian $H_{N,s}$ by

$$H_{N,s}(\sigma) := \frac{\beta\sqrt{s}}{\sqrt{2N}} \sum_{i,j=1}^N g_{ij} \sigma_i \sigma_j + \sum_{i=1}^N (h + \beta\sqrt{(1-s)q} z_i) \sigma_i, \quad (1.7)$$

where (z_i) is a family of independent standard Gaussian random variables. Note that $H_{N,1} = H_N$. We write $\langle \cdot \rangle_s$ for the Gibbs expectation, i.e.

$$\langle F \rangle_s := \frac{1}{Z_{N,s}} \sum_{\sigma \in \Sigma_N} F(\sigma) e^{H_{N,s}(\sigma)}$$

for a function F on Σ_N , where $Z_{N,s} := \sum_{\sigma \in \Sigma_N} e^{H_{N,s}(\sigma)}$. For replicas $\sigma^1, \sigma^2, \dots$ which are sampled independently from the Gibbs measure for fixed disorder, put

$$R_{ab} := \frac{1}{N} \sum_{i=1}^N \sigma_i^a \sigma_i^b.$$

Set

$$\nu_s[f] := \mathbb{E} \langle f \rangle_s, \quad Q_{ab} := R_{ab} - q.$$

We also write

$$\phi_N(s) := \frac{1}{N} \mathbb{E} \log Z_{N,s} \quad (1.8)$$

and define

$$A_s := \nu_s[Q_{12}^2], \quad B_s := \nu_s[Q_{12}Q_{13}], \quad C_s := \nu_s[Q_{12}Q_{34}]. \quad (1.9)$$

Now we state the main theorem of this paper.

Theorem 1.1. *Under the setting above, the overlap concentration holds in the sense that there exists a constant $C_K < \infty$ such that, for every $N \geq 1$,*

$$\sup_{\substack{(\beta, h) \in \mathcal{K} \\ 0 \leq s \leq 1}} N \nu_s[Q_{12}^2] \leq C_K. \quad (1.10)$$

Consequently, the expected free energy

$$\phi_N(\beta, h) := \phi_N(1) = \frac{1}{N} \mathbb{E} \log \sum_{\sigma \in \Sigma_N} e^{H_N(\sigma)}$$

and the replica-symmetric free energy

$$\phi^{\text{RS}}(\beta, h) := \log 2 + \mathbb{E} \log \cosh(h + \beta \sqrt{q} Z) + \frac{\beta^2}{4} (1 - q)^2$$

satisfy the quantitative estimate uniformly on $\mathcal{K}$:

$$0 \leq \phi^{\text{RS}}(\beta, h) - \phi_N(\beta, h) \leq \frac{C_K}{N}. \quad (1.11)$$

Moreover, it holds that

$$N(A_s - 2B_s + C_s) = \frac{\alpha}{\beta^2(1 - s\alpha)} + o_K(1). \quad (1.12)$$

uniformly for $(\beta, h, s) \in \mathcal{K} \times [0, 1]$. Here and below, the notation " $o_K(1) \rightarrow 0$ " as $N \rightarrow \infty$ means the uniform convergence over the compact set $\mathcal{K}$.

Remark that the claim above was previously proved for sufficiently small β , independently by Talagrand, Latała, and the authors; see [19, 7]. Our contribution is to extend this result to the strict de Almeida-Thouless region, which is expected to be the maximal region in which the claim holds.

By taking the limit as $N \rightarrow \infty$ in (1.11) in Theorem 1.1, we recover replica symmetry

$$\lim_{N \rightarrow \infty} \phi_N(\beta, h) = \phi^{\text{RS}}(\beta, h)$$

throughout the entire de Almeida–Thouless region. Establishing replica symmetry in this region had remained an open problem for many years. Very recently, Lopatto [10] resolved this problem by showing that the Parisi formula agrees with the replica-symmetric formula in this region, building on Talagrand’s result that the limiting free energy is given by the Parisi formula [18].

Our main results provide a quantitative counterpart to this qualitative identification. More precisely, instead of asking only whether the thermodynamic limit is replica symmetric or not, we work under the strict AT condition and prove the quantitative estimate, as stated in Theorem 1.1. The strict inequality gives a positive stability margin away from the AT boundary, and our argument turns this margin into finite- N control. Hence, our result complements the characterization of the replica-symmetric phase.

In the following subsection, we briefly review the history of the Sherrington–Kirkpatrick model and the relevant previous work.

1.2 A brief history and previous research

Spin glasses were introduced to describe magnetic materials in which the interactions between microscopic spins are both disordered and frustrated. Experiments on dilute magnetic alloys in the 1970s showed behavior very different from that of ordinary ferromagnets: a sharp freezing transition, very slow relaxation, and a strong dependence on the history of the sample. These results suggested that the low-temperature phase could not be described by a single equilibrium state of the usual kind, and they led to probabilistic models with quenched disorder.

One of the first such models was proposed by Edwards and Anderson [4]. In their model, spins sit on the vertices of a finite-dimensional lattice and interact through random nearest-neighbor couplings. This formulation is physically natural, but its mathematical analysis is extremely difficult. The local interactions compete with each other because of frustration, and the quenched disorder removes the spatial regularity that classical arguments rely on. As a result, even basic questions about the low-temperature phase remain difficult in finite dimensions.

A simplified model was proposed by Sherrington and Kirkpatrick [17]. They replaced the short-range geometry of the Edwards–Anderson model by a mean-field interaction, in which every spin interacts weakly with every other spin. The resulting model, now called the Sherrington–Kirkpatrick (SK) model, keeps both disorder and frustration, but is easier to study by analytic methods. Using the replica method, Sherrington and Kirkpatrick derived a formula for the limiting free energy under the assumption of replica symmetry. The replica method is not mathematically rigorous, because it starts from integer moments of the partition function and then uses an analytic continuation in the number of replicas. More importantly, the replica-symmetric formula gave a negative entropy at low temperature, which is physically inconsistent. It was soon understood that the replica-symmetric ansatz could not describe the whole phase diagram.

A fundamental step was taken by de Almeida and Thouless [3], who studied the local stability of the replica-symmetric solution. Their computation predicted that the replica-symmetric solution is stable when

$$\beta^2 \mathbb{E} \operatorname{sech}^4(h + \beta \sqrt{q} Z) \leq 1, \quad (1.13)$$

and unstable when the reverse strict inequality holds. This boundary is now called the de Almeida–Thouless line, or AT line. Thus, the AT condition comes from the physical stability analysis of the replica-symmetric state.

The failure of replica symmetry below the AT line led Parisi to propose a much richer picture of the low-temperature phase (see [14, 15]). In this picture, the symmetry among replicas is broken in a hierarchical way, and the thermodynamic state is described by an order parameter that gives the distribution of overlaps between typical configurations. This led to the Parisi variational formula for the limiting free energy. The Parisi theory resolved the low-temperature problem of the replica-symmetric solution and gave a single framework that covers both the replica-symmetric and the replica-symmetry-breaking phases.

These ideas have been used far beyond condensed matter physics. In computer science, random constraint satisfaction problems such as random K -SAT and random graph coloring show threshold phenomena similar to replica symmetry breaking, and algorithms based on the cavity method, such as belief and survey propagation, are still among the best known methods for these problems (see [12, 11]). In information theory and high-dimensional statistics, the free energy of suitable spin glass models characterizes the fundamental limits in error-correcting codes, compressed sensing, and inference in graphical models (see [13, 20]). In machine learning, the Hopfield model of associative memory is also a spin glass model in which the stored patterns play the role of quenched disorder (see [6]), and spin glass methods have been used to study loss landscapes and training dynamics of

neural networks. More generally, the notions of frustration, energy landscapes, and metastability, identified in spin glasses, are now standard tools in the study of complex systems, from protein folding to financial networks.

The rigorous mathematical study of the SK model has developed along several different routes. An early landmark was the work of Aizenman, Lebowitz, and Ruelle [1], who proved Gaussian fluctuations of the free energy at high temperature and zero external field by using a cluster expansion. At zero external field, replica symmetry is now known to hold for all $\beta < 1$, and the critical point $\beta = 1$ separates the replica-symmetric and symmetry-breaking regimes. With an external field, the problem turned out to be more difficult. Talagrand proved the replica-symmetric formula at sufficiently high temperature by the cavity method, and a simple interpolation argument due to Latała gave another proof in the range $\beta \in [0, 1/2)$ and $h \in \mathbb{R}$. A recent work of the authors extended Latała's argument to a wider range including the region $\beta \leq 1$ and $h \in \mathbb{R}$ (see [7]).

In parallel, a different line of work addressed the limiting free energy at all temperatures. Guerra [5] introduced an interpolation argument that proved an appropriate upper bound, and then Talagrand [18] proved the matching opposite bound. Together, these results established the Parisi formula for the SK model. This turned the study of the phase diagram of the free energy into a rigorous variational problem.

One question remained: whether the AT stability condition exactly characterizes the replica-symmetric phase of the SK model with a nonzero external field or not. Several rigorous results proved replica symmetry in regions inside the predicted AT region, or replica symmetry breaking on the other side of the line, but reaching the AT boundary itself remained difficult and had been a major open problem in this field. Very recently, Lopatto [10] proved that replica symmetry holds throughout the AT region (1.13) using the variational characterization of the Parisi formula. The present paper gives a quantitative counterpart of this picture (see Theorem 1.1 and the explanation after that).

1.3 Outline of the proof: from coupled free energy to optimal overlap bound

We briefly sketch the proof of the first claim in the main result. The second and third claims are obtained from this using a standard interpolation argument (see Section 4).

Our proof is inspired by that of the Parisi formula in [18]. For $u \in \mathbb{R}$, define the coupled two-replica pressure by

$$p_{N,s}^{(2)}(u) := \frac{1}{2N} \mathbb{E} \log \sum_{\sigma^1, \sigma^2} \exp \left(H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2) + \frac{uN}{2} Q_{12}^2 \right). \quad (1.14)$$

The Guerra–Talagrand interpolation gives

$$p_{N,s}^{(2)}(u) \leq \mathfrak{P}_s(u) + o_K(1) \quad (1.15)$$

uniformly in $(\beta, h) \in \mathcal{K}$, $s \in [0, 1]$, and $u \in [-1, 1]$. Here $\mathfrak{P}_s(u)$ denotes the corresponding one-step replica-symmetry-breaking formula. Using an argument similar to that of Lopatto's work [10], one obtains the flatness condition that for $u > 0$ small enough,

$$\mathfrak{P}_s(u) = \mathfrak{P}_s(0). \quad (1.16)$$

Let

$$D_N(s) := \mathfrak{P}_s(0) - \phi_N(s).$$

By computations using Gaussian integration by parts, we have for $u_0 > 0$ sufficiently small,

$$D'_N(s) = \frac{\beta^2}{4} \nu_s[Q_{12}^2] \leq \frac{\beta^2}{u_0} (p_{N,s}^{(2)}(u_0) - \phi_N(s)) \leq \frac{\beta^2}{u_0} D_N(s) + o_{\mathcal{K}}(1),$$

where we have used Jensen's inequality with $p_{N,s}^{(2)}(u_0) - \phi_N(s) = (2N)^{-1} \mathbb{E} \log \langle e^{u_0 N Q_{12}^2/2} \rangle_s$ for the first inequality, and (1.15) and (1.16) for the last inequality. Since $D_N(0) = 0$, by Gronwall's inequality, we have

$$D_N(s) = o_{\mathcal{K}}(1), \quad p_{N,s}^{(2)}(u_0) - \phi_N(s) = \frac{1}{2N} \mathbb{E} \log \langle e^{u_0 N Q_{12}^2/2} \rangle_s = o_{\mathcal{K}}(1).$$

For a fixed $\delta > 0$, we have

$$\nu_s(|Q_{12}| \geq \delta) \leq \mathbb{P}(\langle e^{u_0 N Q_{12}^2/2} \rangle_s \geq e^{\delta^2 u_0 N/4}) + \mathbb{E} \left[\mathbf{1}\{\langle e^{u_0 N Q_{12}^2/2} \rangle_s < e^{\delta^2 u_0 N/4}\} \langle \mathbf{1}\{|Q_{12}| \geq \delta\} \rangle_s \right].$$

For the first term, using a standard Gaussian Lipschitz concentration inequality for $\log \langle e^{u_0 N Q_{12}^2/2} \rangle_s$, one gets

$$\mathbb{P}(\langle e^{u_0 N Q_{12}^2/2} \rangle_s \geq e^{\delta^2 u_0 N/4}) \leq C_{\mathcal{K},\delta} \exp(-c_{\mathcal{K},\delta} N).$$

For the second term, by Markov's inequality, we have

$$\begin{aligned} & \mathbb{E} \left[\mathbf{1}\{\langle e^{u_0 N Q_{12}^2/2} \rangle_s < e^{\delta^2 u_0 N/4}\} \langle \mathbf{1}\{|Q_{12}| \geq \delta\} \rangle_s \right] \\ & \leq \mathbb{E} \left[\mathbf{1}\{\langle e^{u_0 N Q_{12}^2/2} \rangle_s < e^{\delta^2 u_0 N/4}\} \langle e^{u_0 N Q_{12}^2/2} \rangle_s e^{-u_0 \delta^2 N/2} \right] \\ & \leq C_{\mathcal{K},\delta} \exp(-c_{\mathcal{K},\delta} N). \end{aligned}$$

Therefore, we obtain a weak concentration for the overlap:

$$\nu_s(|Q_{12}| \geq \delta) \leq 2C_{\mathcal{K},\delta} \exp(-c_{\mathcal{K},\delta} N). \quad (1.17)$$

To obtain the N^{-1} bound, we use the cavity bootstrap developed by Talagrand in his high-temperature analysis (see [19, Chapter 1]). Estimating the influence of adding one spin, we reach

$$\nu_s[Q_{12}^2] \leq C_{\mathcal{K}} (\nu_s|Q_{12}|^3 + N^{-1}). \quad (1.18)$$

The weak concentration estimate (1.17) yields that for any fixed $\delta > 0$,

$$\nu_s|Q_{12}|^3 \leq \delta \nu_s[Q_{12}^2] + 8\nu_s(|Q_{12}| \geq \delta) \leq \delta \nu_s[Q_{12}^2] + C_{\mathcal{K},\delta} e^{-c_{\mathcal{K},\delta} N}. \quad (1.19)$$

Consequently, we have

$$\nu_s[Q_{12}^2] \leq \delta C_{\mathcal{K}} \nu_s[Q_{12}^2] + C_{\mathcal{K}} N^{-1} + C_{\mathcal{K}} C_{\mathcal{K},\delta} e^{-c_{\mathcal{K},\delta} N}.$$

Choosing $\delta = 1/(2C_{\mathcal{K}}) > 0$, we obtain

$$\nu_s[Q_{12}^2] \leq \frac{C'_{\mathcal{K}}}{N}, \quad (1.20)$$

with some $C'_{\mathcal{K}}$. This is the sketch of the proof of the quantitative bound for overlaps.

1.4 Organization of the present paper

The rest of the paper is devoted to the proof of Theorem 1.1. In Section 2, we establish coercivity estimates for the replica-symmetric solution. The argument in this part is inspired by Talagrand's approach in [18], and involves several explicit technical calculations. Section 3 gives estimates on overlap concentration. This part is similar to the argument in [7]. In Section 4 we prove the main theorem (Theorem 1.1) by applying the results in Sections 2 and 3.

In the appendices, we prove several technical results used in Sections 2 and 3. In Section A, we prove Proposition 3.4, which provides the overlap estimates used in Section 3. Section B is about the finite-volume Guerra-Talagrand upper bound, which is applied in Section 2.2. In Section C we state and prove Price's theorem, which is also used in Section 2.2.

We also note that we provide a formalization of the proof in Lean 4 [8].

We use $C_{\mathcal{K}}$ and $c_{\mathcal{K}}$ to denote, respectively, large and small positive constants depending only on the compact set $\mathcal{K}$. Their values may vary from line to line.

2 Replica-symmetric coercivity

2.1 Estimates on the replica-symmetric path

Fix $(\beta, h, s) \in \mathcal{K} \times [0, 1]$. For $t \geq 0$, let $\mathbf{H}_t$ be the heat semigroup, i.e.

$$(\mathbf{H}_t \varphi)(x) := \mathbb{E} \varphi(x + \sqrt{t} Z),$$

where Z is a standard normal random variable. Then, it holds that

$$\partial_t (\mathbf{H}_t \varphi)(x) = \frac{1}{2} \mathbf{H}_t (\partial^2 \varphi)(x). \quad (2.1)$$

We introduce the tilted heat semigroup

$$(\mathbf{T}_t \varphi)(x) := e^{-t/2} \frac{(\mathbf{H}_t(\varphi \cdot \cosh))(x)}{\cosh(x)}. \quad (2.2)$$

Differentiation under the Gaussian integral shows that its generator is $\frac{1}{2} \partial_{xx} + \tanh(x) \partial_x$. For each $u \in [0, 1]$, let $X_{s,u}$ denote a random variable whose law is specified by

$$\mathbb{E} \varphi(X_{s,u}) := \begin{cases} (\mathbf{H}_{\beta^2(1-s)q+s\beta^2u} \varphi)(h), & 0 \leq u \leq q, \\ (\mathbf{H}_{\beta^2q} (\mathbf{T}_{s\beta^2(u-q)} \varphi))(h), & q \leq u \leq 1. \end{cases} \quad (2.3)$$

In particular, it holds that

$$X_{s,q} \stackrel{\text{d}}{=} h + \beta \sqrt{q} Z, \quad (2.4)$$

where " $\stackrel{\text{d}}{=}$ " means the equality in law.

Lemma 2.1. *If $s\beta^2(1-q) > 1$ and $q \leq u \leq 1$, then*

$$\beta^2 \mathbb{E} \operatorname{sech}^4(X_{s,u}) \leq \alpha. \quad (2.5)$$

Proof. We first show that

$$h < \sigma^2 := \beta^2 q. \quad (2.6)$$

Indeed, suppose that $h \geq \sigma^2$. A direct calculation shows that

$$1 - q = \mathbb{E} \operatorname{sech}^2(h + \sigma Z) \leq \mathbb{E} \operatorname{sech}^2(\sigma^2 + \sigma Z).$$

Let ε be a symmetric Rademacher variable, independent of Z , and put $Y = \sigma^2 \varepsilon + \sigma Z$. Then $\varepsilon^2 = 1$ and $\mathbb{E}[\varepsilon | Y] = \tanh Y$ due to $\mathbb{P}(\varepsilon = \pm 1, Y \in dy) = \frac{1}{2\sqrt{2\pi\sigma^2}} e^{-(y - (\pm\sigma^2))^2/2\sigma^2} dy$, and hence

$$\mathbb{E} \operatorname{sech}^2(\sigma^2 + \sigma Z) = \mathbb{E} \operatorname{Var}(\varepsilon | Y) \leq \mathbb{E} \left(\varepsilon - \frac{1}{1 + \sigma^2} Y \right)^2 = \frac{1}{1 + \sigma^2}.$$

Therefore

$$(1 - q)(1 + \beta^2 q) \leq 1.$$

Reorganizing this with $q > 0$ implies $\beta^2(1 - q) \leq 1$. This contradicts the assumption $s\beta^2(1 - q) > 1$ because $s \leq 1$. Thus, we obtain (2.6).

Fix $q \leq u \leq 1$ and set

$$t := s\beta^2(u - q), \quad X := h + \sigma Z.$$

From (2.3) it follows that

$$\mathbb{E} \operatorname{sech}^4(X_{s,u}) = \mathbb{E}[(\mathbb{T}_t \operatorname{sech}^4)(X)].$$

Hence, for (2.5) it suffices to prove

$$\mathbb{E}[(\mathbb{T}_t \operatorname{sech}^4)(X)] \leq \mathbb{E} \operatorname{sech}^4(X).$$

Now put

$$g_t := \mathbb{T}_t \operatorname{sech}^4 \quad \text{and} \quad M(t) := \mathbb{E} g_t(X).$$

Since

$$g_t(x) = e^{-t/2} \frac{\mathbf{H}_t(\operatorname{sech}^3)(x)}{\cosh x},$$

g_t is an even and nonincreasing function on $[0, \infty)$. In particular, it holds that

$$g'_t(x) \leq 0, \quad x > 0.$$

By (2.1) and (2.2) we have

$$\partial_t(\mathbb{T}_t \varphi)(x) = \frac{1}{2 \cosh^2 x} \partial_x (\cosh^2 x \partial_x (\mathbb{T}_t \varphi)(x)). \quad (2.7)$$

Let $\phi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ be the standard normal density. Since $X \sim N(h, \sigma^2)$, its density is $\sigma^{-1} \phi((x - h)/\sigma)$. From (2.7) and integration by parts in divergence form we have

$$\begin{aligned} M'(t) &= \frac{1}{2} \int_{\mathbb{R}} \partial_x (\cosh^2 x g'_t(x)) \frac{\phi((x - h)/\sigma)}{\sigma \cosh^2 x} dx \\ &= -\frac{1}{2} \int_{\mathbb{R}} \cosh^2 x g'_t(x) \partial_x \left(\frac{\phi((x - h)/\sigma)}{\sigma \cosh^2 x} \right) dx \\ &= \mathbb{E} \left[\left(\tanh X + \frac{X - h}{2\sigma^2} \right) g'_t(X) \right]. \end{aligned}$$

Pairing x and $-x$ gives

$$M'(t) = \int_0^\infty g'_t(x) K(x) dx, \quad (2.8)$$

where

$$K(x) := \frac{1}{\sigma} \phi\left(\frac{x-h}{\sigma}\right) \left(\tanh x + \frac{x-h}{2\sigma^2}\right) + \frac{1}{\sigma} \phi\left(\frac{-x-h}{\sigma}\right) \left(\tanh x + \frac{x+h}{2\sigma^2}\right).$$

Since

$$\frac{\phi((-x-h)/\sigma)}{\phi((x-h)/\sigma)} = e^{-2hx/\sigma^2},$$

we obtain

$$\frac{\sigma K(x)}{\phi((x-h)/\sigma) + \phi((-x-h)/\sigma)} = \tanh x + \frac{x}{2\sigma^2} - \frac{h}{2\sigma^2} \tanh\left(\frac{hx}{\sigma^2}\right).$$

Since $0 < h/\sigma^2 < 1$ due to (2.6), $\tanh\left(\frac{hx}{\sigma^2}\right) \leq \tanh x$. This inequality implies that for $x > 0$

$$\frac{\sigma K(x)}{\phi((x-h)/\sigma) + \phi((-x-h)/\sigma)} \geq \left(1 - \frac{h}{2\sigma^2}\right) \tanh x + \frac{x}{2\sigma^2} > 0.$$

Thus $K(x) > 0$, while $g'_t(x) \leq 0$, and therefore (2.8) yields

$$M'(t) \leq 0.$$

In particular, it holds that $M(t) \leq M(0)$. From this inequality we obtain

$$\beta^2 \mathbb{E} \operatorname{sech}^4(X_{s,u}) \leq \beta^2 \mathbb{E} \operatorname{sech}^4(h + \beta\sqrt{q} Z) = \alpha.$$

□

For $0 \leq u \leq 1$ and $x \in \mathbb{R}$, define

$$\Psi(u, x) := \mathbb{E} \log \cosh\left(x + \beta\sqrt{s(q-u)_+} Z\right) + \frac{s\beta^2}{2} (1 - \max\{u, q\}), \quad Z \sim N(0, 1), \quad (2.9)$$

where $(a)_+ := \max\{a, 0\}$. Equivalently, we can define $\Psi(u, x)$ by

$$\Psi(u, x) = \begin{cases} (\mathbb{H}_{s\beta^2(q-u)} \log \cosh)(x) + \frac{s\beta^2}{2} (1 - q), & 0 \leq u \leq q, \\ \log \cosh(x) + \frac{s\beta^2}{2} (1 - u), & q \leq u \leq 1. \end{cases}$$

Differentiating (2.9) in x gives

$$\partial_x \Psi(u, x) = \mathbb{E} \tanh\left(x + \beta\sqrt{s(q-u)_+} Z\right) = \begin{cases} (\mathbb{H}_{s\beta^2(q-u)} \tanh)(x), & 0 \leq u \leq q, \\ \tanh(x), & q \leq u \leq 1. \end{cases} \quad (2.10)$$

The replica-symmetric free energy for $H_{N,s}$ along the path is defined by

$$P_s^* := \log 2 + \mathbb{E} \Psi(0, h + \beta\sqrt{(1-s)q} Z) - \frac{s\beta^2}{2} \int_q^1 u \, du \quad (2.11)$$

$$= \log 2 + \mathbb{E} \log \cosh(h + \beta\sqrt{q} Z) + \frac{s\beta^2}{4} (1 - q)^2, \quad (2.12)$$

where we have used the semigroup property and $\beta^2(1-s)q + s\beta^2q = \beta^2q$.

Proposition 2.2. *The function g_s defined by*

$$g_s(u) := \mathbb{E}[(\partial_x \Psi(u, X_{s,u}))^2] \quad (2.13)$$

satisfies

$$g_s(u) - u \begin{cases} > 0, & 0 \leq u < q, \\ = 0, & u = q, \\ < 0, & q < u \leq 1. \end{cases} \quad (2.14)$$

Moreover, there are constants $c_K, \varepsilon_K > 0$ such that

$$|g_s(u) - u| \geq c_K |u - q| \quad \text{whenever } |u - q| \leq \varepsilon_K. \quad (2.15)$$

Proof. Since (2.10) and (2.4) imply

$$g_s(q) = \mathbb{E}[\tanh^2(X_{s,q})] = q,$$

(2.15) holds for $u = q$.

We next consider the case $u < q$. Since (2.10) gives

$$g_s(u) = \mathbb{E}[(H_{s\beta^2(q-u)}(\tanh))(X_{s,u})^2],$$

by noting that (2.1) and (2.3) imply

$$\partial_u \mathbb{E}\varphi(X_{s,u}) = \frac{s\beta^2}{2} (H_{\beta^2(1-s)q+s\beta^2u}(\partial^2\varphi))(h) = \frac{s\beta^2}{2} \mathbb{E}[(\partial^2\varphi)(X_{s,u})],$$

we have

$$\begin{aligned} g'_s(u) &= -s\beta^2 \partial_v \mathbb{E}[(H_v(\tanh))(X_{s,u})^2] \big|_{v=s\beta^2(q-u)} + \partial_v \mathbb{E}[(H_{s\beta^2(q-u)}(\tanh))(X_{s,v})^2] \big|_{v=u} \\ &= -s\beta^2 \mathbb{E}[(H_{s\beta^2(q-u)}(\tanh))(X_{s,u})(H_{s\beta^2(q-u)}(\partial^2 \tanh))(X_{s,u})] \\ &\quad + \frac{s\beta^2}{2} \mathbb{E}\left[\left(\partial_x^2 (H_{s\beta^2(q-u)}(\tanh)(x))^2\right) \big|_{x=X_{s,u}}\right] \\ &= s\beta^2 \mathbb{E}[(H_{s\beta^2(q-u)}(\partial \tanh))(X_{s,u})^2]. \end{aligned}$$

Thus, we obtain

$$g'_s(u) = s\beta^2 \mathbb{E}\left[\{(H_{s\beta^2(q-u)}(\text{sech}^2))(X_{s,u})\}^2\right].$$

Jensen's inequality, the semigroup property and (2.3) imply

$$g'_s(u) \leq s\beta^2 \mathbb{E}[(H_{s\beta^2(q-u)}(\text{sech}^4))(X_{s,u})] = s\beta^2 \mathbb{E} \text{sech}^4(h + \beta\sqrt{q}Z) = s\alpha.$$

Since $g_s(q) = q$, the integration of both sides implies

$$q - g_s(u) \leq s\alpha(q - u).$$

Hence, it holds that

$$g_s(u) - u \geq (1 - s\alpha)(q - u) \quad (0 \leq u < q). \quad (2.16)$$

This yields (2.14) for $0 \leq u < q$.

We consider the case $u \geq q$. In this case, (2.10) gives $g_s(u) = \mathbb{E} \tanh^2(X_{s,u})$. On the other hand, (2.1), (2.2) and (2.3) imply

$$\begin{aligned} \partial_u \mathbb{E} \varphi(X_{s,u}) &= \partial_u H_{\beta^2 q} \left(e^{-\frac{s\beta^2}{2}(u-q)} \frac{H_{s\beta^2(u-q)}(\varphi \cdot \cosh)}{\cosh} \right) (h) \\ &= -\frac{s\beta^2}{2} \mathbb{E} \varphi(X_{s,u}) + \frac{s\beta^2}{2} H_{\beta^2 q} \left(e^{-\frac{s\beta^2}{2}(u-q)} \frac{H_{s\beta^2(u-q)}(\partial^2(\varphi \cdot \cosh))}{\cosh} \right) (h) \\ &= \frac{s\beta^2}{2} \mathbb{E}(\partial^2 \varphi)(X_{s,u}) + s\beta^2 H_{\beta^2 q} \left(e^{-\frac{s\beta^2}{2}(u-q)} \frac{H_{s\beta^2(u-q)}((\partial \varphi) \cdot \sinh)}{\cosh} \right) (h). \end{aligned}$$

Hence, it holds that

$$\begin{aligned} g'_s(u) &= s\beta^2 \mathbb{E} \operatorname{sech}^4(X_{s,u}) - 2s\beta^2 \mathbb{E} \tanh^2(X_{s,u}) \operatorname{sech}^2(X_{s,u}) \\ &\quad + 2s\beta^2 H_{\beta^2 q} \left(e^{-\frac{s\beta^2}{2}(u-q)} \frac{H_{s\beta^2(u-q)}(\operatorname{sech} \cdot \tanh^2)}{\cosh} \right) (h). \end{aligned}$$

Thus, by recalling (2.2) and (2.3), we have

$$g'_s(u) = s\beta^2 \mathbb{E} \operatorname{sech}^4(X_{s,u}), \quad g'_s(q) = s\alpha < 1. \quad (2.17)$$

If $s\beta^2(1-q) > 1$, Lemma 2.1 gives $g'_s(u) \leq s\alpha < 1$, so $g_s(u) < u$ for $u > q$. We consider the case $s\beta^2(1-q) \leq 1$. Since $\operatorname{sech}^4(x) \leq \operatorname{sech}^2(x) = 1 - \tanh^2(x)$ for every $x \in \mathbb{R}$, by (2.17) we have

$$g'_s(u) \leq s\beta^2(1 - g_s(u)). \quad (2.18)$$

Then, (2.18) implies that for $u > q$

$$\partial_u(g_s(u) - u) \leq -s\beta^2(g_s(u) - u) + s\beta^2(1 - u) - 1 \leq -s\beta^2(g_s(u) - u).$$

Hence, by noting that (2.17) implies $g_s(u) - u < 0$ for $u > q$ sufficiently close to q and by applying Grönwall's inequality we have

$$g_s(u) - u < 0 \quad (u > q).$$

This yields (2.14) for $u > q$.

To prove (2.15), note first that (2.16) and (1.5) give

$$g_s(u) - u \geq \delta_{\mathcal{K}}(q - u), \quad 0 \leq u < q.$$

Moreover, the expression for $g'_s(u)$ in (2.17) is jointly continuous in (β, h, s, u) on the set $q \leq u \leq 1$. Since $g'_s(q) = s\alpha \leq 1 - \delta_{\mathcal{K}}$, compactness gives $\varepsilon_{\mathcal{K}} > 0$ such that whenever $q \leq u \leq q + \varepsilon_{\mathcal{K}}$,

$$g'_s(u) \leq 1 - \frac{\delta_{\mathcal{K}}}{2}.$$

Thus, for $q < u \leq q + \varepsilon_{\mathcal{K}}$,

$$g_s(u) - u = \int_q^u (g'_s(v) - 1) \, dv \leq -\frac{\delta_{\mathcal{K}}}{2}(u - q).$$

Putting things together, we have (2.15) with $c_{\mathcal{K}} = \delta_{\mathcal{K}}/2$ after decreasing $\varepsilon_{\mathcal{K}}$ if necessary. $\square$

2.2 Two-replica Guerra–Talagrand coercivity

For $\lambda \in \mathbb{R}$, define a function f_λ on $\mathbb{R}^2$ by

$$f_\lambda(x_1, x_2) := \log \left(\frac{1}{4} \sum_{\varepsilon_1, \varepsilon_2 = \pm 1} \exp(\varepsilon_1 x_1 + \varepsilon_2 x_2 + \lambda \varepsilon_1 \varepsilon_2) \right). \quad (2.19)$$

For $(\beta, h, s) \in \mathcal{K} \times [0, 1]$, define a function ξ_s on $\mathbb{R}$ by

$$\xi_s(r) := \beta^2(1-s)qr + \frac{s\beta^2}{2}r^2. \quad (2.20)$$

Given $v \in [-1, 1]$, put $\iota_v := 1$ for $v \geq 0$ and $\iota_v := -1$ for $v < 0$. Let $\{a_j\}_{j=0}^M$ be the increasing list of the distinct elements of $\{0, q, |v|, 1\}$, i.e.

$$\{a_j\}_{j=0}^M = \{0, q, |v|, 1\} \quad \text{and} \quad a_j < a_{j+1} \quad (0 \leq j \leq M-1).$$

Note that $M = 2$ if $|v| \in \{0, q, 1\}$, and $M = 3$ otherwise. Define

$$Q_j^v := \begin{pmatrix} a_j & \iota_v(a_j \wedge |v|) \\ \iota_v(a_j \wedge |v|) & a_j \end{pmatrix}, \quad 0 \leq j \leq M. \quad (2.21)$$

For $0 \leq j \leq M$, set

$$m_j := \begin{cases} 0, & a_j \leq q, \\ \frac{1}{2}, & q < a_j \leq |v|, \\ 1, & a_j > \max\{q, |v|\}. \end{cases} \quad (2.22)$$

We call $(a_j)_{j=0}^M$ the breakpoints and $(m_j)_{j=0}^M$ the levels. For each v , the breakpoints and levels are given as follows.

	the breakpoints $(a_j)_{j=0}^M$	the levels $(m_j)_{j=0}^M$
$ v = 0$	$0, q, 1$	$0, 0, 1$
$0 < v < q$	$0, v , q, 1$	$0, 0, 0, 1$
$ v = q$	$0, q, 1$	$0, 0, 1$
$q < v < 1$	$0, q, v , 1$	$0, 0, \frac{1}{2}, 1$
$ v = 1$	$0, q, 1$	$0, 0, \frac{1}{2}$

We also set

$$m_{M+1} = 1.$$

Define a 2×2 -matrix B_j by

$$B_j := \beta^2(1-s)q \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + s\beta^2 Q_j^v = \xi'_s(Q_j^v), \quad 0 \leq j \leq M. \quad (2.23)$$

Since $|v|$ is a breakpoint, every increment is of one of the forms

$$Q_j^v - Q_{j-1}^v = \begin{cases} (a_j - a_{j-1}) \begin{pmatrix} 1 & \iota_v \\ \iota_v & 1 \end{pmatrix}, & a_j \leq |v|, \\ (a_j - a_{j-1})I_2, & a_j > |v|. \end{cases} \quad (2.24)$$

Here, note that $a_{j-1} \geq |v|$ if $a_j > |v|$. Both displayed matrices are positive semidefinite. Consequently,

$$B_j - B_{j-1} = s\beta^2(Q_j^v - Q_{j-1}^v) \succeq 0. \quad (2.25)$$

For a positive semidefinite 2×2 matrix C , let G_C be a centered Gaussian vector with covariance C , and define

$$\mathcal{T}_{m,C}F(x) := \begin{cases} m^{-1} \log \mathbb{E} e^{mF(x+G_C)}, & m > 0, \\ \mathbb{E} F(x+G_C), & m = 0. \end{cases} \quad (2.26)$$

Define functions $U_{s,M}^{\lambda,v}$ by

$$U_{s,M}^{\lambda,v} = f_\lambda, \quad U_{s,j-1}^{\lambda,v} = \mathcal{T}_{m_j, B_j - B_{j-1}} U_{s,j}^{\lambda,v}, \quad (1 \leq j \leq M), \quad (2.27)$$

and for $x = (x_1, x_2) \in \mathbb{R}^2$ set

$$U_s^{\lambda,v}(x) := U_{s,0}^{\lambda,v}(x). \quad (2.28)$$

At a breakpoint a_j , we use the notation

$$U_s^{\lambda,v}(a_j, x) := U_{s,j}^{\lambda,v}(x). \quad (2.29)$$

We give another construction of $U_s^{\lambda,v}$ via random variables. Choose independent centered Gaussian vectors $G_1, \dots, G_M$ with $\text{Var}(G_j) = B_j - B_{j-1}$ and let $\mathcal{F}_j = \sigma(G_1, \dots, G_j)$, with $\mathcal{F}_0$ trivial. For fixed $x \in \mathbb{R}^2$, put

$$X_M^\lambda = f_\lambda \left(x + \sum_{k=1}^M G_k \right),$$

and for $0 \leq j \leq M$ define X_j recursively by

$$X_{j-1}^\lambda := \begin{cases} \frac{1}{m_j} \log \mathbb{E} \left[e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1} \right], & m_j > 0, \\ \mathbb{E} \left[X_j^\lambda \mid \mathcal{F}_{j-1} \right], & m_j = 0. \end{cases} \quad (2.30)$$

Then, it holds that

$$X_j^\lambda = U_{s,j}^{\lambda,v} \left(x + \sum_{k=1}^j G_k \right), \quad X_0^\lambda = U_s^{\lambda,v}(x). \quad (2.31)$$

In particular, $U_s^{\lambda,v}(x)$ is deterministic. This definition also covers the case $s = 0$. In this case, every covariance increment vanishes.

Next, put

$$d_j := \frac{s\beta^2}{2} \sum_{k,\ell=1}^2 (Q_{j,k\ell}^v)^2, \quad 0 \leq j \leq M, \quad (2.32)$$

where $Q_{j,k\ell}^v$ is the (k, ℓ) entry of Q_j^v . As a function of a breakpoint value a , this quantity is given by

$$d(a) = \begin{cases} 2s\beta^2 a^2, & 0 \leq a \leq |v|, \\ s\beta^2(a^2 + |v|^2), & |v| \leq a \leq 1. \end{cases}$$

Indeed, it holds that $d_j = d(a_j)$. Since $d(a)$ is continuous and nondecreasing,

$$d_j - d_{j-1} \geq 0. \quad (2.33)$$

Define

$$\mathcal{L}_s(v) := \frac{1}{2} \sum_{j=1}^M m_j(d_j - d_{j-1}). \quad (2.34)$$

This normalization gives the scalar Gaussian contribution at the endpoint $t = 1$ of the interpolation in Appendix B, and is independent of v . Indeed, if $|v| \leq q$, then

$$\sum_{j=1}^M m_j(d_j - d_{j-1}) = d(1) - d(q) = s\beta^2(1 - q^2).$$

If $q < |v| < 1$,

$$\begin{aligned} \sum_{j=1}^M m_j(d_j - d_{j-1}) &= \frac{1}{2}(d(|v|) - d(q)) + d(1) - d(|v|) \\ &= s\beta^2(|v|^2 - q^2) + s\beta^2(1 - |v|^2) = s\beta^2(1 - q^2). \end{aligned}$$

If $|v| = 1$,

$$\sum_{j=1}^M m_j(d_j - d_{j-1}) = \frac{1}{2}(d(1) - d(q)) = s\beta^2(1 - q^2).$$

Therefore,

$$\mathcal{L}_s(v) = \frac{s\beta^2}{2}(1 - q^2), \quad (2.35)$$

In particular, this yields that $\mathcal{L}_s(v)$ is independent of v .

With $Y_0 = h + \beta\sqrt{(1-s)q}Z$ where Z is a standard normal random variable, set

$$\mathfrak{P}_s(\lambda, v) := 2\log 2 + \mathbb{E}U_s^{\lambda, v}(Y_0, Y_0) - \lambda v - \mathcal{L}_s(v). \quad (2.36)$$

Recall the notation P_s^* from (2.12). For $v \in \mathcal{R}_N := \{-1, -1 + 2/N, \dots, 1\}$, define

$$Z_{N,s}^{(2)}(v) := \sum_{\substack{\sigma^1, \sigma^2 \in \Sigma_N \\ R_{12}=v}} e^{H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2)}. \quad (2.37)$$

The following finite-volume Guerra–Talagrand upper bound follows from Lemma B.2, which is proved in Appendix B. It is a special case of the general two-dimensional bound obtained by Talagrand and Chen in [19, 2]:

$$\frac{1}{N} \mathbb{E} \log Z_{N,s}^{(2)}(v) \leq \inf_{\lambda \in \mathbb{R}} \mathfrak{P}_s(\lambda, v). \quad (2.38)$$

Lemma 2.3. *In the recursion (2.30), define the tilted conditional expectation by*

$$\mathbb{E}_j^\lambda[H] := \frac{\mathbb{E} \left[H e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1} \right]}{\mathbb{E} \left[e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1} \right]}, \quad m_j > 0,$$

and let $\mathbb{E}_j^\lambda[H] = \mathbb{E}[H \mid \mathcal{F}_{j-1}]$ when $m_j = 0$. Then

$$\partial_\lambda X_{j-1}^\lambda = \mathbb{E}_j^\lambda \left[\partial_\lambda X_j^\lambda \right], \quad (2.39)$$

$$\partial_{\lambda\lambda} X_{j-1}^\lambda = \mathbb{E}_j^\lambda \left[\partial_{\lambda\lambda} X_j^\lambda \right] + m_j \text{Var}_j^\lambda \left(\partial_\lambda X_j^\lambda \right). \quad (2.40)$$

Proof. For $m_j > 0$, recalling that

$$X_{j-1}^\lambda = \frac{1}{m_j} \log \mathbb{E} \left[e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1} \right],$$

we have

$$\begin{aligned} \partial_\lambda X_{j-1}^\lambda &= \frac{\mathbb{E} \left[\partial_\lambda X_j^\lambda e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1} \right]}{\mathbb{E} \left[e^{m_j X_j^\lambda} \mid \mathcal{F}_{j-1} \right]} = \mathbb{E}_j^\lambda [\partial_\lambda X_j^\lambda], \\ \partial_{\lambda\lambda} X_{j-1}^\lambda &= \mathbb{E}_j^\lambda [\partial_{\lambda\lambda} X_j^\lambda] + m_j \text{Var}_j^\lambda (\partial_\lambda X_j^\lambda). \end{aligned}$$

For $m_j = 0$, direct differentiation of the ordinary conditional expectation gives the same identities with no variance term. This completes the proof. $\square$

Corollary 2.4. *It holds that*

$$\left| \partial_\lambda U_s^{\lambda,v}(x) \right| \leq 1, \quad 0 \leq \partial_{\lambda\lambda} U_s^{\lambda,v}(x) \leq \frac{5}{2}. \quad (2.41)$$

Consequently, uniformly in $(\beta, h, s, v, \lambda)$,

$$0 \leq \partial_{\lambda\lambda} \mathfrak{P}_s(\lambda, v) \leq \frac{5}{2}. \quad (2.42)$$

Proof. Let $\langle \cdot \rangle_{\lambda,x}$ denote expectation on $\{\pm 1\}^2$ with weights proportional to

$$\exp(\varepsilon_1 x_1 + \varepsilon_2 x_2 + \lambda \varepsilon_1 \varepsilon_2) \quad ((\varepsilon_1, \varepsilon_2) \in \{\pm 1\}^2).$$

Since $(\varepsilon_1 \varepsilon_2)^2 = 1$,

$$\partial_\lambda f_\lambda(x) = \langle \varepsilon_1 \varepsilon_2 \rangle_{\lambda,x}, \quad \partial_{\lambda\lambda} f_\lambda(x) = 1 - \langle \varepsilon_1 \varepsilon_2 \rangle_{\lambda,x}^2.$$

Hence, we have $|\partial_\lambda f_\lambda| \leq 1$ and $0 \leq \partial_{\lambda\lambda} f_\lambda \leq 1$. By applying (2.39) recursively, we have

$$\text{Var}_j^\lambda (\partial_\lambda X_j^\lambda) \leq \mathbb{E}_j^\lambda [(\partial_\lambda X_j^\lambda)^2] \leq 1.$$

Similarly, applying (2.40) recursively, we have

$$\partial_{\lambda\lambda} X_0^\lambda = \mathbb{E}_1^\lambda \cdots \mathbb{E}_M^\lambda \left[\partial_{\lambda\lambda} X_M^\lambda \right] + \sum_{j=1}^M m_j \mathbb{E}_1^\lambda \cdots \mathbb{E}_{j-1}^\lambda \left[\text{Var}_j^\lambda (\partial_\lambda X_j^\lambda) \right].$$

This equality, (2.31) and the fact that $0 \leq \partial_{\lambda\lambda} f_\lambda \leq 1$ yield

$$0 \leq \partial_{\lambda\lambda} U_s^{\lambda,v}(x) \leq 1 + \sum_{j=1}^M m_j. \quad (2.43)$$

If $|v| \leq q$, the sum is at most 1. If $q < |v| < 1$, $(m_j)_{j=1}^M = (0, \frac{1}{2}, 1)$ and their sum is 3/2. If $|v| = 1$, $(m_j)_{j=1}^M = (0, \frac{1}{2})$ and their sum is 1/2. Consequently, we have in every case,

$$1 + \sum_{j=1}^M m_j \leq \frac{5}{2}.$$

Together with (2.43), this proves (2.41).

Moreover, differentiation under the expectation in (2.36) gives

$$\partial_{\lambda\lambda} \mathfrak{P}_s(\lambda, v) = \mathbb{E} \partial_{\lambda\lambda} U_s^{\lambda,v}(Y_0, Y_0).$$

Hence (2.41) proves (2.42). $\square$

Proposition 2.5. *Put*

$$\Gamma_s(v) := \inf_{\lambda \in \mathbb{R}} \mathfrak{P}_s(\lambda, v).$$

There is a constant $c_K > 0$ such that for $(\beta, h, s, v) \in \mathcal{K} \times [0, 1] \times [-1, 1]$,

$$\Gamma_s(v) \leq 2P_s^* - c_K(v - q)^2. \quad (2.44)$$

Moreover,

$$\Gamma_s(q) = 2P_s^*. \quad (2.45)$$

Proof. We first evaluate the finite recursion at $\lambda = 0$. Suppose that $-q \leq v \leq q$. In this case, $a_{M-1} = q$, $m_M = 1$ and $B_M - B_{M-1} = s\beta^2(1 - q)I_2$. Hence, with independent Gaussian random variables G_1 and G_2 with means 0 and variances $s\beta^2(1 - q)$, the identities

$$f_0(x_1, x_2) = \log \cosh x_1 + \log \cosh x_2, \quad \partial_\lambda f_\lambda(x_1, x_2)|_{\lambda=0} = \tanh x_1 \tanh x_2,$$

(2.26), (2.27) and (2.29) give

$$\begin{aligned} U_s^{0,v}(q, x_1, x_2) &= \log \mathbb{E} \left[\frac{1}{4} \sum_{\varepsilon_1, \varepsilon_2 \in \pm 1} \exp(\varepsilon_1(x_1 + G_1) + \varepsilon_2(x_2 + G_2)) \right] \\ &= s\beta^2(1 - q) + \log \cosh x_1 + \log \cosh x_2, \end{aligned}$$

and

$$\begin{aligned} \partial_\lambda U_s^{\lambda,v}(q, x_1, x_2) \Big|_{\lambda=0} &= \frac{\mathbb{E} \left[\frac{1}{4} \sum_{\varepsilon_1, \varepsilon_2 \in \pm 1} \varepsilon_1 \varepsilon_2 \exp(\varepsilon_1(x_1 + G_1) + \varepsilon_2(x_2 + G_2)) \right]}{\mathbb{E} \left[\frac{1}{4} \sum_{\varepsilon_1, \varepsilon_2 \in \pm 1} \exp(\varepsilon_1(x_1 + G_1) + \varepsilon_2(x_2 + G_2)) \right]} \\ &= \tanh x_1 \tanh x_2. \end{aligned}$$

Hence, by these equalities, the fact that $m_j = 0$ for $j < M$, and (2.24), we obtain

$$U_s^{0,v}(x_1, x_2) = s\beta^2(1 - q) + \mathbb{E} \log \cosh(x_1 + \beta\sqrt{s q} Z_0) + \mathbb{E} \log \cosh(x_2 + \beta\sqrt{s q} Z_0), \quad (2.46)$$

$$\begin{aligned} \partial_\lambda U_s^{\lambda,v}(x_1, x_2) \Big|_{\lambda=0} &= \mathbb{E} \left[\tanh(x_1 + \beta\sqrt{s|v|} Z_0 + \beta\sqrt{s(q - |v|)} Z_1) \right. \\ &\quad \left. \times \tanh(x_2 + \beta\sqrt{s|v|} Z_0 + \beta\sqrt{s(q - |v|)} Z_2) \right], \end{aligned} \quad (2.47)$$

where Z_0 , Z_1 and Z_2 are independent standard normal random variables. From these equalities it follows that

$$\begin{aligned} \mathbb{E} U_s^{0,v}(Y_0, Y_0) &= s\beta^2(1 - q) + 2\mathbb{E} \log \cosh(h + \beta\sqrt{q} Z) \\ \partial_\lambda \mathbb{E} U_s^{\lambda,v}(Y_0, Y_0) \Big|_{\lambda=0} &= \mathbb{E} [\tanh(Y_1(v)) \tanh(Y_2(v))] =: \tilde{g}_s(v), \end{aligned}$$

where $Y_0 = h + \beta\sqrt{(1 - s)q} Z$, Z is a standard normal random variable independent from other random variables, and $(Y_1(v), Y_2(v))$ is a Gaussian pair such that both $Y_1(v)$ and $Y_2(v)$ have means h and variances $\beta^2 q$, and have the covariance

$$\text{Cov}(Y_1(v), Y_2(v)) = \beta^2((1 - s)q + sv).$$

Thus, by applying (2.12) and (2.35) we have

$$\mathfrak{P}_s(0, v) = 2P_s^*, \quad \partial_\lambda \mathfrak{P}_s(0, v) = \tilde{g}_s(v) - v, \quad -q \leq v \leq q. \quad (2.48)$$

Since Price's theorem (Theorem C.1) implies

$$\tilde{g}'_s(v) = s\beta^2 \mathbb{E}[\operatorname{sech}^2 Y_1(v) \operatorname{sech}^2 Y_2(v)],$$

the Cauchy–Schwarz inequality yields

$$\tilde{g}'_s(v) \leq s\beta^2 \mathbb{E}[\operatorname{sech}^4(h + \beta\sqrt{q}Z)] \leq s\alpha,$$

where α is defined in (1.3). Since $\operatorname{Cov}(Y_1(q), Y_2(q)) = \beta^2 q = \operatorname{Var}(Y_1(q))$, it holds that $\tilde{g}_s(q) = \mathbb{E}[\tanh^2(Y_1(q))] = q$. Hence, it follows that

$$\tilde{g}_s(v) - v \geq (1 - s\alpha)(q - v) > 0, \quad -q \leq v < q. \quad (2.49)$$

In the case $0 \leq v \leq q$, put $t = s\beta^2(q - v)$, let X be a Gaussian random variable with mean h and variance $\beta^2((1 - s)q + sv)$ which is independent of other random variables. The variables

$$X + \sqrt{t} Z_1, \quad X + \sqrt{t} Z_2,$$

have the joint law of $(Y_1(v), Y_2(v))$, and are independent under the probability measure conditioned on X . Hence, it follows that

$$\tilde{g}_s(v) = \mathbb{E} \left[\left(H_t(\tanh)(X) \right)^2 \right] = g_s(v), \quad 0 \leq v \leq q \quad (2.50)$$

in view of (2.3), (2.10), and (2.13).

Now we consider the case $q < v \leq 1$. Let $t := s\beta^2(v - q)$. Similarly to the case $-q \leq v \leq q$, it holds that for $q < v < 1$

$$\begin{aligned} U_s^{0,v}(v, x_1, x_2) &= s\beta^2(1 - v) + \log \cosh x_1 + \log \cosh x_2, \\ \partial_\lambda U_s^{\lambda,v}(v, x_1, x_2) \Big|_{\lambda=0} &= \tanh x_1 \tanh x_2. \end{aligned}$$

Since $a_{M-1} = v$, $a_{M-2} = q$, $m_{M-1} = \frac{1}{2}$ and

$$B_{M-1} - B_{M-2} = s\beta^2(v - q) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = t \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

we have

$$U_s^{0,v}(q, x_1, x_2) = s\beta^2(1 - v) + 2 \log \mathbb{E} \sqrt{\cosh(x_1 + \sqrt{t} Z_0) \cosh(x_2 + \sqrt{t} Z_0)},$$

and

$$\begin{aligned} & \partial_\lambda U_s^{\lambda,v}(q, x_1, x_2) \Big|_{\lambda=0} \\ &= \frac{\mathbb{E} \left[\partial_\lambda U_s^{\lambda,v}(v, x_1 + \sqrt{t} Z_0, x_2 + \sqrt{t} Z_0) \Big|_{\lambda=0} \exp \left(\frac{1}{2} U_s^{0,v}(v, x_1 + \sqrt{t} Z_0, x_2 + \sqrt{t} Z_0) \right) \right]}{\mathbb{E} \exp \left(\frac{1}{2} U_s^{0,v}(v, x_1 + \sqrt{t} Z_0, x_2 + \sqrt{t} Z_0) \right)} \\ &= \frac{\mathbb{E} \left[\tanh(x_1 + \sqrt{t} Z_0) \tanh(x_2 + \sqrt{t} Z_0) \sqrt{\cosh(x_1 + \sqrt{t} Z_0) \cosh(x_2 + \sqrt{t} Z_0)} \right]}{\mathbb{E} \left[\sqrt{\cosh(x_1 + \sqrt{t} Z_0) \cosh(x_2 + \sqrt{t} Z_0)} \right]}. \end{aligned}$$

Note that these equalities also hold for $v = 1$. Hence, by letting $x_1 = x_2$, we obtain

$$U_s^{0,v}(q, x_1, x_1) = 2 \log \cosh x_1 + s\beta^2(1 - q), \quad (2.51)$$

$$\partial_\lambda U_s^{\lambda,v}(q, x_1, x_1) \Big|_{\lambda=0} = \frac{\mathbb{E}[\tanh^2(x_1 + \sqrt{t} Z_0) \cosh(x_1 + \sqrt{t} Z_0)]}{\mathbb{E} \cosh(x_1 + \sqrt{t} Z_0)} = (\mathbb{T}_t \tanh^2)(x_1), \quad (2.52)$$

where we applied the fact that $\mathbb{E} \cosh(x + \sqrt{t} Z) = e^{t/2} \cosh x$. Since $a_1 = q$, $m_1 = 0$ and $B_1 - B_0 = s\beta^2 q \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, from (2.51) and (2.52) it follows that

$$\begin{aligned} U_s^{0,v}(x_1, x_1) &= 2\mathbb{E} \log \cosh(x_1 + \beta\sqrt{sq} Z_0) + s\beta^2(1 - q), \\ \partial_\lambda U_s^{\lambda,v}(x_1, x_1) \Big|_{\lambda=0} &= \mathbb{E}(\mathbb{T}_t \tanh^2)(x_1 + \beta\sqrt{sq} Z_0). \end{aligned}$$

Since these equalities yield

$$\begin{aligned} \mathbb{E} U_s^{0,v}(Y_0, Y_0) &= 2\mathbb{E} \log \cosh(h + \beta\sqrt{q} Z_0) + s\beta^2(1 - q), \\ \mathbb{E} \partial_\lambda U_s^{\lambda,v}(Y_0, Y_0) \Big|_{\lambda=0} &= (\mathbb{H}_{\beta^2 q} \mathbb{T}_{s\beta^2(v-q)} \tanh^2)(h) = g_s(v), \end{aligned}$$

where (2.10), (2.3), and (2.13) are applied in the last equality. Thus, together with (2.48) and (2.50), we obtain

$$\mathfrak{P}_s(0, v) = 2P_s^*, \quad \partial_\lambda \mathfrak{P}_s(0, v) = g_s(v) - v, \quad 0 \leq v \leq 1. \quad (2.53)$$

We also need a strict gap in the case $v < -q$. Suppose first that $s > 0$ and let $t = s\beta^2(|v| - q) > 0$. Again, similarly to the case $-q \leq v \leq q$, it holds that for $v < -q$

$$U_s^{0,v}(|v|, x_1, x_2) = s\beta^2(1 - |v|) + \log \cosh x_1 + \log \cosh x_2.$$

Hence, since $|v| > q$, an argument similar to the one above gives

$$U_s^{0,v}(q, x_1, x_2) = s\beta^2(1 - |v|) + 2 \log \mathbb{E} \sqrt{\cosh(x_1 + \sqrt{t} Z) \cosh(x_2 - \sqrt{t} Z)}.$$

From this equality, the Cauchy–Schwarz inequality and the fact that $\mathbb{E} \cosh(x + \sqrt{t} Z) = e^{t/2} \cosh x$, it follows that

$$\begin{aligned} U_s^{0,v}(q, x_1, x_2) &\leq s\beta^2(1 - |v|) + t + \log \cosh x_1 + \log \cosh x_2 \\ &= \log \cosh x_1 + \log \cosh x_2 + s\beta^2(1 - q). \end{aligned}$$

Since $t > 0$, equality holds only when $x_1 = -x_2$. From the fact that $a_1 = q$ and $m_1 = 0$ we have

$$U_s^{0,v}(x_1, x_2) \leq \mathbb{E} \log \cosh(x_1 + \beta\sqrt{sq} Z_0) + \mathbb{E} \log \cosh(x_2 - \beta\sqrt{sq} Z_0) + s\beta^2(1 - q).$$

Since

$$\mathbb{P}[Y_0 + \beta\sqrt{sq} Z_0 = -(Y_0 - \beta\sqrt{sq} Z_0)] = \mathbb{P}(Y_0 = 0) = 0,$$

where $Y_0 = h + \beta\sqrt{(1-s)q}Z$ and $h > 0$, we have the strict inequality:

$$\mathbb{E} U_s^{0,v}(Y_0, Y_0) < 2\mathbb{E} \log \cosh(h + \beta\sqrt{q} Z) + s\beta^2(1 - q).$$

This yields

$$\mathfrak{P}_s(0, v) < 2P_s^*, \quad v < -q, \quad s > 0. \quad (2.54)$$

When $s = 0$, a direct calculation gives

$$\mathfrak{P}_0(0, v) = 2P_0^*, \quad \partial_\lambda \mathfrak{P}_0(0, v) = q - v, \quad v < -q. \quad (2.55)$$

We now prove the desired bound by applying the results above. Choose $\varepsilon_{\mathcal{K}} < q_{\mathcal{K}}/2$ so that (2.15) holds when $|v - q| \leq \varepsilon_{\mathcal{K}}$. Proposition 2.2 gives $a_{\mathcal{K}} \in (0, 1)$ such that

$$(g_s(v) - v)(v - q) \leq -a_{\mathcal{K}}(v - q)^2, \quad |v - q| \leq \varepsilon_{\mathcal{K}}. \quad (2.56)$$

Since $v \geq 0$ for $|v - q| \leq \varepsilon_{\mathcal{K}}$ with sufficiently small $\varepsilon_{\mathcal{K}} > 0$, (2.42) and (2.53) imply

$$\mathfrak{P}_s(\lambda, v) \leq 2P_s^* + \lambda(g_s(v) - v) + \frac{5}{4}\lambda^2.$$

Taking $\lambda = 2a_{\mathcal{K}}(v - q)/5 \in [-1, 1]$ and applying (2.56) to this inequality yield

$$\Gamma_s(v) \leq 2P_s^* - \frac{a_{\mathcal{K}}^2}{5}(v - q)^2, \quad |v - q| \leq \varepsilon_{\mathcal{K}}. \quad (2.57)$$

It remains to consider overlaps bounded away from q . We claim that

$$\inf_{\lambda \in [-1, 1]} \mathfrak{P}_s(\lambda, v) < 2P_s^* \quad \text{for } v \neq q. \quad (2.58)$$

In the case $0 \leq v < q$ and $q < v \leq 1$, this follows from (2.14) and (2.53) by moving $\lambda \in (0, 1]$ slightly in the direction opposite to the derivative. For the case $-q \leq v < 0$, use (2.48) and (2.49). For the case $v < -q$ and $s > 0$, apply (2.54). For the case $v < -q$ and $s = 0$, (2.55) gives the claim by taking a sufficiently small negative $\lambda \in [-1, 0)$.

Due to the compactness of $\mathcal{K}$, there exists $\delta_{\mathcal{K}} > 0$ such that

$$\Gamma_s(v) \leq \inf_{\lambda \in [-1, 1]} \mathfrak{P}_s(\lambda, v) \leq 2P_s^* - \delta_{\mathcal{K}}, \quad |v - q| \geq \varepsilon_{\mathcal{K}}.$$

Since $|v - q| \leq 2$,

$$\Gamma_s(v) \leq 2P_s^* - \frac{\delta_{\mathcal{K}}}{4}(v - q)^2, \quad |v - q| \geq \varepsilon_{\mathcal{K}}. \quad (2.59)$$

Combining (2.57) and (2.59) proves (2.44), with

$$c_{\mathcal{K}} = \min \left\{ \frac{a_{\mathcal{K}}^2}{5}, \frac{\delta_{\mathcal{K}}}{4} \right\} > 0.$$

Finally, at $v = q$, (2.53) gives

$$\mathfrak{P}_s(0, q) = 2P_s^*, \quad \partial_\lambda \mathfrak{P}_s(0, q) = g_s(q) - q = 0.$$

Since (2.42) implies that $\lambda \mapsto \mathfrak{P}_s(\lambda, q)$ is convex, $\lambda = 0$ is a global minimizer. Hence

$$\Gamma_s(q) = 2P_s^*,$$

which proves (2.45). □

Together with (2.38), the proposition above yields the following.

Corollary 2.6. *For every N and $v \in \mathcal{R}_N$,*

$$\frac{1}{N} \mathbb{E} \log Z_{N,s}^{(2)}(v) \leq 2P_s^* - c_{\mathcal{K}}(v - q)^2. \quad (2.60)$$

3 Preliminary overlap concentration

3.1 Gaussian calculus

We record Gaussian integration by parts, which is used below. Let $G := (G_1, \dots, G_m)$ be centered Gaussian with covariance matrix C . If F is continuously differentiable with bounded first derivatives, then Gaussian integration by parts (see [19, Vol. 1, Sec. A.4]) gives

$$\mathbb{E}[G_i F(G)] = \sum_{j=1}^m C_{ij} \mathbb{E}[\partial_j F(G)]. \quad (3.1)$$

We also use the following standard consequence of Gaussian concentration inequality. Let G be a standard Gaussian vector, and let $\{F_v\}_{v \in I}$ be a finite set of L -Lipschitz functions on the underlying vector space. Then, it holds that

$$\mathbb{E} \max_{v \in I} \{F_v(G) - \mathbb{E} F_v(G)\} \leq L \sqrt{2 \log |I|}. \quad (3.2)$$

Indeed, the Gaussian concentration inequality (see, for example, [9]) gives, for every $t > 0$ and every $v \in I$,

$$\mathbb{E} \exp(t(F_v(G) - \mathbb{E} F_v(G))) \leq \exp\left(\frac{t^2 L^2}{2}\right).$$

This inequality and Jensen's inequality imply

$$\begin{aligned} \mathbb{E} \max_{v \in I} (F_v(G) - \mathbb{E} F_v(G)) &\leq \frac{1}{t} \log \mathbb{E} \exp\left(t \max_{v \in I} (F_v(G) - \mathbb{E} F_v(G))\right) \\ &\leq \frac{1}{t} \log \sum_{v \in I} \mathbb{E} \exp(t(F_v(G) - \mathbb{E} F_v(G))) \\ &\leq \frac{\log |I|}{t} + \frac{t L^2}{2}, \end{aligned}$$

where the first inequality follows from Jensen's inequality. For $|I| \geq 2$ and $L > 0$, optimize the upper bound by choosing $t = \sqrt{2 \log |I|}/L$. The cases $|I| = 1$ and $L = 0$ are trivial. Therefore, we obtain (3.2).

3.2 Finite-volume Gronwall closure

For $\rho \geq 0$, define the coupled pressure $p_{N,s}^{(2)}(\rho)$ and its normalized excess $F_{N,s}(\rho)$ by

$$p_{N,s}^{(2)}(\rho) := \frac{1}{2N} \mathbb{E} \log \sum_{\sigma^1, \sigma^2} \exp\left(H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2) + \frac{\rho N}{2} Q_{12}^2\right), \quad (3.3)$$

$$F_{N,s}(\rho) := p_{N,s}^{(2)}(\rho) - \phi_N(s) = \frac{1}{2N} \mathbb{E} \log \left\langle e^{\rho N Q_{12}^2/2} \right\rangle_s, \quad (3.4)$$

respectively. Fix

$$0 < \rho_0 < 2c_K, \quad (3.5)$$

where c_K is the coercivity constant in Proposition 2.5.

Lemma 3.1. *There is $C_K > 0$ such that for any $s \in [0, 1]$ and $N \in \mathbb{N}$,*

$$p_{N,s}^{(2)}(\rho_0) \leq P_s^* + C_K \sqrt{\frac{\log(N+1)}{N}}. \quad (3.6)$$

Proof. Let $L_v := \log Z_{N,s}^{(2)}(v)$. Each L_v is a Lipschitz function of the Gaussian disorder with squared Lipschitz constant at most $O_K(1)N$. Indeed,

$$|\partial_{g_{ij}} L_v| \leq \frac{\sqrt{2}\beta\sqrt{s}}{\sqrt{N}}, \quad |\partial_{z_i} L_v| \leq 2\beta\sqrt{(1-s)q},$$

and summing up the squares over i, j and i gives the claim. Hence, by (3.2),

$$\mathbb{E} \max_{v \in \mathcal{R}_N} (L_v - \mathbb{E} L_v) \leq C_K \sqrt{N \log(N+1)}, \quad (3.7)$$

where we recall $\mathcal{R}_N := \{-1, -1 + 2/N, \dots, 1\}$. Thus, by using (2.60), (3.5), and (3.7), we have

$$\begin{aligned} 2Np_{N,s}^{(2)}(\rho_0) &= \mathbb{E} \log \sum_{v \in \mathcal{R}_N} \exp \left(\mathbb{E} L_v + (L_v - \mathbb{E} L_v) + \frac{\rho_0 N}{2} (v - q)^2 \right) \\ &\leq \log(N+1) + \max_v \left\{ \mathbb{E} L_v + \frac{\rho_0 N}{2} (v - q)^2 \right\} + C_K \sqrt{N \log(N+1)} \\ &\leq 2NP_s^* + C_K \sqrt{N \log(N+1)}. \end{aligned}$$

This proves (3.6). $\square$

Put

$$D_N(s) := P_s^* - \phi_N(s). \quad (3.8)$$

Lemma 3.2. *There is $C_K > 0$ such that for $s \in [0, 1]$*

$$D_N(s) \leq C_K \sqrt{\frac{\log(N+1)}{N}}, \quad A_s = \nu_s[Q_{12}^2] \leq C_K \sqrt{\frac{\log(N+1)}{N}}. \quad (3.9)$$

Proof. Write

$$\epsilon_N := C_K \sqrt{\frac{\log(N+1)}{N}}.$$

Similarly to [7, (14)], via Gaussian integration by parts (see [19, Vol. 1, Sec. A.4]), it follows that

$$\phi'_N(s) = \frac{\beta^2}{4} \left((1-q)^2 - \nu_s[Q_{12}^2] \right). \quad (3.10)$$

Combining this identity with (2.12) and the definition of A_s in (1.9) yields

$$D'_N(s) = \frac{\beta^2}{4} A_s. \quad (3.11)$$

On the other hand, convexity of $F_{N,s}(\rho)$ in $\rho \in [0, \infty)$, where $F_{N,s}(\rho)$ is defined in (3.4), gives

$$\frac{1}{4} A_s = \partial_\rho F_{N,s}(\rho)|_{\rho=0+} \leq \frac{F_{N,s}(\rho_0)}{\rho_0}. \quad (3.12)$$

By (3.6),

$$F_{N,s}(\rho_0) \leq D_N(s) + \epsilon_N. \quad (3.13)$$

Consequently,

$$D'_N(s) \leq \frac{\beta^2}{\rho_0} (D_N(s) + \epsilon_N). \quad (3.14)$$

Gronwall's inequality, the compactness of $\mathcal{K}$, the continuity of $D_N(s)$ in $s \in [0, 1]$, and the fact that $D_N(0) = 0$ imply the inequality for $D_N(s)$ in (3.9). The estimate for A_s in (3.9) follows from this inequality, (3.12) and (3.13). $\square$

3.3 Fixed-deviation overlap concentration

Corollary 3.3. *For every $\varepsilon > 0$, there are constants $c_{\mathcal{K},\varepsilon}, C_{\mathcal{K},\varepsilon} > 0$ such that*

$$\sup_{\substack{(\beta,h) \in \mathcal{K} \\ 0 \leq s \leq 1}} \nu_s[|Q_{12}| \geq \varepsilon] \leq C_{\mathcal{K},\varepsilon} e^{-c_{\mathcal{K},\varepsilon} N}. \quad (3.15)$$

Proof. Let

$$Y_{N,s} := \log \left\langle e^{\rho_0 N Q_{12}^2 / 2} \right\rangle_s.$$

By (3.4), (3.9) and (3.13),

$$\mathbb{E} Y_{N,s} = 2N F_{N,s}(\rho_0) \leq C_{\mathcal{K}} N \epsilon_N = o(N) \quad (3.16)$$

uniformly in $s \in [0, 1]$. It is known that the map

$$\begin{aligned} ((g_{ij})_{i,j}, (Z_i)_i) &\mapsto Y_{N,s} = \log \sum_{\sigma^1, \sigma^2} \exp \left(H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2) + \frac{\rho_0 N}{2} Q_{12}^2 \right) \\ &\quad - \log \sum_{\sigma^1, \sigma^2} \exp (H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2)), \end{aligned}$$

where we note that $H_{N,s}$ depends on $((g_{ij})_{i,j}, (Z_i)_i)$, is Lipschitz continuous with Lipschitz constant $C_{\mathcal{K}} \sqrt{N}$ (see the proof of Lemma 3.1 or [19, Theorem 1.3.5] and the discussion after the theorem). The Gaussian concentration inequality (see [19, Theorem 1.3.4]) therefore gives, for all large enough N ,

$$\mathbb{P}(|Y_{N,s} - \mathbb{E} Y_{N,s}| > t) \leq \exp(-N^{-1} t^2 / C_{\mathcal{K}}) \quad (t \in [0, \infty)).$$

This inequality and (3.16) yield

$$\mathbb{P}\left(Y_{N,s} > \frac{\rho_0 \varepsilon^2 N}{4}\right) \leq C_{\mathcal{K},\varepsilon} e^{-c_{\mathcal{K},\varepsilon} N}. \quad (3.17)$$

For each realization of the disorder, on $\{|Q_{12}| \geq \varepsilon\}$ we have $\mathbf{1}_{\{|Q_{12}| \geq \varepsilon\}} \leq \exp\left(\frac{\rho_0 N}{2}(Q_{12}^2 - \varepsilon^2)\right)$. Therefore,

$$\langle \mathbf{1}_{\{|Q_{12}| \geq \varepsilon\}} \rangle_s \leq \exp\left(-\frac{\rho_0 \varepsilon^2 N}{2}\right) \left\langle \exp\left(\frac{\rho_0 N Q_{12}^2}{2}\right) \right\rangle_s = \exp\left(-\frac{\rho_0 \varepsilon^2 N}{2} + Y_{N,s}\right).$$

Hence, (3.17) implies

$$\begin{aligned} \nu_s[|Q_{12}| \geq \varepsilon] &\leq \mathbb{E} \left[\exp\left(-\frac{\rho_0 \varepsilon^2 N}{2} + Y_{N,s}\right) \mathbf{1}_{\{Y_{N,s} \leq \frac{\rho_0 \varepsilon^2 N}{4}\}} \right] + \mathbb{P}\left(Y_{N,s} > \frac{\rho_0 \varepsilon^2 N}{4}\right) \\ &\leq \exp\left(-\frac{\rho_0 \varepsilon^2 N}{4}\right) + \mathbb{P}\left(Y_{N,s} > \frac{\rho_0 \varepsilon^2 N}{4}\right). \end{aligned}$$

Therefore, we obtain (3.15). $\square$

3.4 Cavity closure

Recall from (1.9) that

$$A_s = \nu_s[Q_{12}^2], \quad B_s = \nu_s[Q_{12}Q_{13}], \quad C_s = \nu_s[Q_{12}Q_{34}].$$

Introduce the three cavity modes

$$U_s := A_s - 4B_s + 3C_s, \quad V_s := 2B_s - 3C_s, \quad D_s := A_s - 2B_s + C_s, \quad (3.18)$$

and put

$$\kappa := 1 - 4q + 3r, \quad \zeta := 2q + q^2 - 3r. \quad (3.19)$$

Proposition 3.4. *The last-spin cavity expansion has a common three-edge closure, as follows.*

For the scalar specialization, there are remainders $\mathcal{R}_{N,s}^U, \mathcal{R}_{N,s}^V, \mathcal{R}_{N,s}^D$ and $C_K > 0$ such that

$$(1 - s\beta^2\kappa)U_s = \frac{\kappa}{N} + \mathcal{R}_{N,s}^U, \quad (3.20)$$

$$(1 - s\beta^2\kappa)V_s - s\beta^2\zeta U_s = \frac{\zeta}{N} + \mathcal{R}_{N,s}^V, \quad (3.21)$$

$$(1 - s\alpha)D_s = \frac{\alpha}{\beta^2 N} + \mathcal{R}_{N,s}^D, \quad (3.22)$$

and uniformly on $\mathcal{K} \times [0, 1]$,

$$\max_{X \in \{U, V, D\}} |\mathcal{R}_{N,s}^X| \leq C_K(N^{-3/2} + \nu_s[|Q_{12}|^3]). \quad (3.23)$$

More generally, let a bounded cavity observable distinguish a marked replica edge only according as a second edge is the same edge, shares one endpoint with it, or is disjoint from it. At the decoupled endpoint the three corresponding last-spin coefficients are

$$\theta_A = 1 - q^2, \quad \theta_B = q - q^2, \quad \theta_C = r - q^2. \quad (3.24)$$

If T_A, T_B, T_C are the resulting same-edge, shared-endpoint, and disjoint-edge correlations, the first cavity derivative is governed by the edge-incidence matrix

$$\mathcal{M}_s = s\beta^2 \begin{pmatrix} \theta_A & -4\theta_B & 3\theta_C \\ \theta_B & \theta_A - 2\theta_B - 3\theta_C & 6\theta_C - 3\theta_B \\ \theta_C & 4\theta_B - 8\theta_C & \theta_A - 8\theta_B + 10\theta_C \end{pmatrix}. \quad (3.25)$$

For a vector $T = (T_A, T_B, T_C)^\top$, write

$$\mathcal{U}(T) = T_A - 4T_B + 3T_C, \quad \mathcal{V}(T) = 2T_B - 3T_C, \quad \mathcal{D}(T) = T_A - 2T_B + T_C.$$

The same change of variables as in (3.18) gives

$$\mathcal{U}(\mathcal{M}_s T) = s\beta^2 \kappa \mathcal{U}(T), \quad \mathcal{V}(\mathcal{M}_s T) = s\beta^2 (\zeta \mathcal{U}(T) + \kappa \mathcal{V}(T)), \quad \mathcal{D}(\mathcal{M}_s T) = s\alpha \mathcal{D}(T). \quad (3.26)$$

Whenever the second cavity derivative and the replacement of the decoupled correlations by their endpoint values are $O(\eta_N)$, this first-order expansion has an $O(\eta_N)$ remainder in each of the three mode equations.

The scalar specialization follows from a straightforward, but technical, application of Talagrand's cavity argument (see [19, Chapter 1]); its details are given in Appendix A. For the general statement, apply the derivative identity (A.3). The decoupled last-spin expectation is θ_A , θ_B , or θ_C according to the three incidence classes, by (A.9). Enumerating the edges in that identity gives (3.25). Direct multiplication gives (3.26); explicitly,

$$\theta_A - 4\theta_B + 3\theta_C = \kappa, \quad 2\theta_B - 3\theta_C = \zeta, \quad \theta_A - 2\theta_B + \theta_C = 1 - 2q + r = \frac{\alpha}{\beta^2}.$$

Taylor's formula for the cavity interpolation proves the final assertion.

Since $r \leq q$,

$$\alpha - \beta^2\kappa = 2\beta^2(q - r) \geq 0.$$

Thus $\beta^2\kappa \leq \alpha < 1$. If $\kappa < 0$, then $1 - s\beta^2\kappa \geq 1$; if $\kappa \geq 0$, then $1 - s\beta^2\kappa \geq 1 - \alpha$. Hence (1.5) gives

$$1 - s\alpha \geq \delta_{\mathcal{K}}, \quad 1 - s\beta^2\kappa \geq \delta_{\mathcal{K}} \quad (3.27)$$

for all $(\beta, h, s) \in \mathcal{K} \times [0, 1]$. Equations (3.20)–(3.22), in this order, and the boundedness of ζ on $\mathcal{K}$ imply

$$|U_s| + |V_s| + |D_s| \leq C_{\mathcal{K}} \left(\frac{1}{N} + N^{-3/2} + \nu_s[|Q_{12}|^3] \right).$$

Since $A_s = -V_s - 2U_s + 3D_s$, we conclude that

$$A_s \leq \frac{C_{\mathcal{K}}}{N} + C_{\mathcal{K}}\nu_s[|Q_{12}|^3]. \quad (3.28)$$

4 Proof of the main theorem

4.1 Absorption of the cubic remainder and the proof of (1.10)

Fix $\varepsilon > 0$. Since $|Q_{12}| \leq 2$,

$$\begin{aligned} \nu_s[|Q_{12}|^3] &\leq \varepsilon \nu_s[Q_{12}^2] + 8\nu_s[|Q_{12}| > \varepsilon] \\ &\leq \varepsilon A_s + 8C_{\mathcal{K},\varepsilon} e^{-c_{\mathcal{K},\varepsilon}N}, \end{aligned} \quad (4.1)$$

where Corollary 3.3 was used. Choose $\varepsilon = \varepsilon_{\mathcal{K}} = 1/(2C_{\mathcal{K}})$ so that the coefficient of A_s obtained by substituting (4.1) into (3.28) is at most $1/2$. We obtain

$$A_s = \nu_s[Q_{12}^2] \leq \frac{C'_{\mathcal{K}}}{N},$$

with some $C'_{\mathcal{K}} > 0$ uniformly on $\mathcal{K} \times [0, 1]$. This proves (1.10)

4.2 Quantitative estimate of the free energy: Proof of (1.11)

Recall that $D_N(s) := P_s^* - \phi_N(s)$. The identity (3.11), $D'_N(s) = \frac{\beta^2}{4}\nu_s[Q_{12}^2]$, with $P_1^* = \phi^{\text{RS}}(\beta, h)$ and $D_N(0) = 0$ gives

$$\phi^{\text{RS}}(\beta, h) - \phi_N(1) = D_N(1) = \frac{\beta^2}{4} \int_0^1 \nu_s[Q_{12}^2] \, ds. \quad (4.2)$$

The integrand is nonnegative, and the bound above implies that the right-hand side is at most $C_{\mathcal{K}}/N$. This proves (1.11).

4.3 Replicon susceptibility: Proof of (1.12)

The $O(N^{-1})$ second-moment estimate and (4.1) imply, for every fixed $\varepsilon > 0$,

$$\sup_{\mathcal{K} \times [0,1]} N \nu_s[|Q_{12}|^3] \leq C_{\mathcal{K}} \varepsilon + 8N C_{\mathcal{K},\varepsilon} e^{-c_{\mathcal{K},\varepsilon} N}.$$

First let $N \rightarrow \infty$ and then $\varepsilon \downarrow 0$. Thus

$$\nu_s[|Q_{12}|^3] = o_{\mathcal{K}}(N^{-1})$$

uniformly in s . Equation (3.22), together with (3.23) and (3.27), now gives

$$ND_s = \frac{\alpha}{\beta^2(1-s\alpha)} + o_{\mathcal{K}}(1).$$

This proves (1.12) and completes the proof of Theorem 1.1.

5 Central limit theorem for the overlap

In this section we work at the endpoint $s = 1$ of the smart path and suppress the subscript s . Put

$$a := 1 - 2q + r = \frac{\alpha}{\beta^2}.$$

The argument preceding (3.27) gives $\beta^2 \kappa \leq \alpha < 1$, and hence

$$1 - \alpha > 0, \quad 1 - \beta^2 \kappa > 0. \tag{5.1}$$

Define

$$\sigma^2 := \frac{3a}{1-\alpha} - \frac{2\kappa}{1-\beta^2\kappa} - \frac{\zeta}{(1-\beta^2\kappa)^2}. \tag{5.2}$$

Our second main theorem is the following.

Theorem 5.1 (Central limit theorem for the overlap). *For every fixed (β, h) in the strict Almeida–Thouless region,*

$$\sqrt{N} (R_{12} - q) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

with respect to the disorder-averaged two-replica Gibbs measure, where σ^2 is given by (5.2). Equivalently, for every $t \in \mathbb{R}$,

$$\nu \left[\cos \left(t \sqrt{N} Q_{12} \right) \right] \longrightarrow \exp \left(-\frac{\sigma^2 t^2}{2} \right), \tag{5.3}$$

$$\nu \left[\sin \left(t \sqrt{N} Q_{12} \right) \right] \longrightarrow 0. \tag{5.4}$$

Proof. We divide the computation into several paragraphs.

Uniform second moment along the last-spin interpolation. Let ν_u , $0 \leq u \leq 1$, be the last-spin interpolation from Appendix A, so that $\nu_1 = \nu$. Recall

$$\varepsilon_a = \sigma_N^a, \quad Q_{ab}^- = \frac{1}{N} \sum_{i < N} \sigma_i^a \sigma_i^b - q, \quad Q_{ab} = Q_{ab}^- + \frac{1}{N} \varepsilon_a \varepsilon_b.$$

The optimal overlap estimate already proved in the main text gives

$$\nu[Q_{12}^2] \leq \frac{C}{N}. \quad (5.5)$$

Set

$$M_2(u) := \nu_u[(Q_{12}^-)^2].$$

Applying (A.3) to $F = (Q_{12}^-)^2$, every term in $M_2'(u)$ has the form

$$\nu_u[(Q_{12}^-)^2 \varepsilon_a \varepsilon_b Q_{ab}^-].$$

Since $|Q_{ab}^-| \leq 2$,

$$|\nu_u[(Q_{12}^-)^2 \varepsilon_a \varepsilon_b Q_{ab}^-]| \leq 2M_2(u).$$

The number of terms in the cavity derivative is fixed, and β is fixed, hence

$$|M_2'(u)| \leq CM_2(u). \quad (5.6)$$

Moreover,

$$|Q_{12}^-|^2 \leq 2|Q_{12}|^2 + \frac{2}{N^2},$$

so (5.5) gives

$$M_2(1) \leq \frac{C}{N}.$$

Gronwall's inequality applied backward from $u = 1$ therefore yields

$$\sup_{0 \leq u \leq 1} \nu_u[(Q_{12}^-)^2] \leq \frac{C}{N}. \quad (5.7)$$

By replica symmetry the same estimate holds for every fixed replica edge. Consequently, for all fixed edges ab, cd ,

$$\sup_{0 \leq u \leq 1} \nu_u[|Q_{ab}^-|] \leq \frac{C}{\sqrt{N}}, \quad (5.8)$$

$$\sup_{0 \leq u \leq 1} \nu_u[|Q_{ab}^- Q_{cd}^-|] \leq \frac{C}{N}. \quad (5.9)$$

The second inequality follows from Cauchy–Schwarz.

Three test-function correlations. Let $f \in C^2(\mathbb{R})$ satisfy

$$\|f\|_\infty + \|f'\|_\infty + \|f''\|_\infty < \infty.$$

Put

$$X_{ab} := \sqrt{N} Q_{ab}.$$

We consider the three replica-edge classes

$$T_A(f) := \nu[X_{12}f(X_{12})], \quad (5.10)$$

$$T_B(f) := \nu[X_{13}f(X_{12})], \quad (5.11)$$

$$T_C(f) := \nu[X_{34}f(X_{12})]. \quad (5.12)$$

After the disorder average the sites are exchangeable. Hence, for every edge ab appearing above,

$$\nu[Q_{ab}f(X_{12})] = \nu[(\varepsilon_a\varepsilon_b - q)f(X_{12})].$$

Thus, if

$$\bar{\varepsilon}_{ab} := \varepsilon_a\varepsilon_b - q,$$

then

$$T_E(f) = \sqrt{N} \nu[\bar{\varepsilon}_{e(E)}f(X_{12})], \quad e(A) = 12, \quad e(B) = 13, \quad e(C) = 34. \quad (5.13)$$

For $E \in \{A, B, C\}$ define

$$g_E(u) := \sqrt{N} \nu_u[\bar{\varepsilon}_{e(E)}f(X_{12})].$$

Then

$$g_E(1) = T_E(f).$$

It is convenient to remove the last-spin contribution from the argument of f . Put

$$Y := \sqrt{N} Q_{12}^- + \frac{q}{\sqrt{N}}. \quad (5.14)$$

Since

$$X_{12} = \sqrt{N} Q_{12}^- + \frac{\varepsilon_1\varepsilon_2}{\sqrt{N}},$$

we have the exact identity

$$X_{12} = Y + \frac{\bar{\varepsilon}_{12}}{\sqrt{N}}. \quad (5.15)$$

At $u = 0$ the last spin is independent of the first $N - 1$ spins, and its effective field has the distribution

$$h + \beta\sqrt{q} Z.$$

Therefore, for distinct replica labels,

$$\mathbb{E}_0[\varepsilon_a\varepsilon_b] = q, \quad \mathbb{E}_0[\varepsilon_a\varepsilon_b\varepsilon_c\varepsilon_d] = r.$$

In particular,

$$\mathbb{E}_0[\bar{\varepsilon}_{12}^2] = 1 - q^2, \quad (5.16)$$

$$\mathbb{E}_0[\bar{\varepsilon}_{13}\bar{\varepsilon}_{12}] = q - q^2, \quad (5.17)$$

$$\mathbb{E}_0[\bar{\varepsilon}_{34}\bar{\varepsilon}_{12}] = r - q^2. \quad (5.18)$$

For example,

$$\mathbb{E}_0[\bar{\varepsilon}_{13}\bar{\varepsilon}_{12}] = \mathbb{E}_0[\varepsilon_2\varepsilon_3] - q\mathbb{E}_0[\varepsilon_1\varepsilon_3] - q\mathbb{E}_0[\varepsilon_1\varepsilon_2] + q^2 = q - q^2.$$

By Taylor's theorem and (5.15),

$$f(X_{12}) = f(Y) + \frac{\bar{\varepsilon}_{12}}{\sqrt{N}} f'(Y) + \rho_N,$$

where

$$|\rho_N| \leq \frac{\|f''\|_\infty}{2N} |\bar{\varepsilon}_{12}|^2 \leq \frac{2\|f''\|_\infty}{N}.$$

Because $\mathbb{E}_0[\bar{\varepsilon}_{e(E)}] = 0$, independence at $u = 0$ and (5.16)–(5.18) give

$$g_E(0) = \theta_E \nu_0[f'(Y)] + O_f(N^{-1/2}), \quad (5.19)$$

where

$$\theta_A = 1 - q^2, \quad \theta_B = q - q^2, \quad \theta_C = r - q^2. \quad (5.20)$$

Here and below $O_f(\cdot)$ means that the implicit constant depends only on β, h and $\|f\|_\infty + \|f'\|_\infty + \|f''\|_\infty$.

We next replace $\nu_0[f'(Y)]$ by the endpoint quantity $\nu[f'(X_{12})]$. Applying (A.3) to the bounded cavity observable $f'(Y)$ and using (5.8),

$$\left| \frac{d}{du} \nu_u[f'(Y)] \right| \leq \frac{C\|f'\|_\infty}{\sqrt{N}}.$$

Also, by (5.15),

$$|f'(X_{12}) - f'(Y)| \leq \frac{2\|f''\|_\infty}{\sqrt{N}}.$$

Hence

$$\nu_0[f'(Y)] = \nu[f'(X_{12})] + O_f(N^{-1/2}). \quad (5.21)$$

Combining (5.19) and (5.21),

$$g_E(0) = \theta_E m_f + O_f(N^{-1/2}), \quad m_f := \nu[f'(X_{12})]. \quad (5.22)$$

First cavity derivative and the replica-edge matrix. Define the cavity analogues

$$\tilde{T}_A(f) := \sqrt{N} \nu_0[Q_{12}^- f(Y)], \quad (5.23)$$

$$\tilde{T}_B(f) := \sqrt{N} \nu_0[Q_{13}^- f(Y)], \quad (5.24)$$

$$\tilde{T}_C(f) := \sqrt{N} \nu_0[Q_{34}^- f(Y)]. \quad (5.25)$$

We compute $g'_E(0)$ using (A.3). In the derivative, replacing $f(X_{12})$ by $f(Y)$ costs $O_f(N^{-1/2})$: indeed,

$$|f(X_{12}) - f(Y)| \leq \frac{2\|f'\|_\infty}{\sqrt{N}},$$

and each derivative term contains one cavity overlap, whose first moment is $O(N^{-1/2})$ by (5.8).

Introduce

$$b_2 := \beta^2(1 - q^2), \quad b_1 := \beta^2(q - q^2), \quad b_0 := \beta^2(r - q^2).$$

The three rows of the cavity matrix are obtained as follows.

For $E = A$, use two replicas in (A.3). The positive pair is 12, the two negative pairs are 13, 23, and the final pair is 34. By the edge rule,

$$g'_A(0) = b_2 \tilde{T}_A - 4b_1 \tilde{T}_B + 3b_0 \tilde{T}_C + O_f(N^{-1/2}).$$

For $E = B$, use three replicas. The positive pairs 12, 13, 23 contribute

$$b_1 \tilde{T}_A + (b_2 + b_1) \tilde{T}_B.$$

The three negative terms are multiplied by -3 . The edges 14 and 24 give respectively the coefficients b_1 and b_0 in the shared-index class, while 34 gives the coefficient b_1 in the disjoint class. Finally the coefficient of the new edge 45 is $6b_0$. Therefore

$$\begin{aligned} g'_B(0) &= b_1 \tilde{T}_A + (b_2 + b_1) \tilde{T}_B - 3(b_1 + b_0) \tilde{T}_B - 3b_1 \tilde{T}_C + 6b_0 \tilde{T}_C + O_f(N^{-1/2}) \\ &= b_1 \tilde{T}_A + (b_2 - 2b_1 - 3b_0) \tilde{T}_B + (6b_0 - 3b_1) \tilde{T}_C + O_f(N^{-1/2}). \end{aligned} \quad (5.26)$$

For $E = C$, use four replicas. Among the six positive pairs, the edge 12 is disjoint from 34, the four edges 13, 14, 23, 24 share one endpoint with 34, and the edge 34 is the same edge. Hence the positive contribution is

$$b_0 \tilde{T}_A + 4b_1 \tilde{T}_B + b_2 \tilde{T}_C.$$

For the four negative terms, multiplied by -4 , the edges 15, 25 are disjoint from 34 and belong to the shared-index class relative to 12, whereas 35, 45 share one endpoint with 34 and belong to the disjoint class relative to 12. Thus the negative contribution is

$$-8b_0 \tilde{T}_B - 8b_1 \tilde{T}_C.$$

The final edge 56 is disjoint from 34 and contributes $10b_0 \tilde{T}_C$. Therefore

$$g'_C(0) = b_0 \tilde{T}_A + (4b_1 - 8b_0) \tilde{T}_B + (b_2 - 8b_1 + 10b_0) \tilde{T}_C + O_f(N^{-1/2}). \quad (5.27)$$

Thus, with

$$T(f) = \begin{pmatrix} T_A(f) \\ T_B(f) \\ T_C(f) \end{pmatrix}, \quad \tilde{T}(f) = \begin{pmatrix} \tilde{T}_A(f) \\ \tilde{T}_B(f) \\ \tilde{T}_C(f) \end{pmatrix}, \quad \theta = \begin{pmatrix} 1 - q^2 \\ q - q^2 \\ r - q^2 \end{pmatrix},$$

and

$$M = \begin{pmatrix} b_2 & -4b_1 & 3b_0 \\ b_1 & b_2 - 2b_1 - 3b_0 & 6b_0 - 3b_1 \\ b_0 & 4b_1 - 8b_0 & b_2 - 8b_1 + 10b_0 \end{pmatrix}, \quad (5.28)$$

equations (5)–(5.27) become

$$g'(0) = M \tilde{T}(f) + O_f(N^{-1/2}). \quad (5.29)$$

Cavity Taylor remainder and endpoint replacement. Apply the cavity derivative formula once more. Every term in $g''_E(u)$ is a fixed numerical multiple of an expectation containing two cavity overlaps. Since

$$\left| \sqrt{N} \bar{\varepsilon}_{e(E)} f(X_{12}) \right| \leq 2\sqrt{N} \|f\|_\infty,$$

equation (5.9) gives

$$\sup_{0 \leq u \leq 1} |g''_E(u)| \leq \frac{C_f}{\sqrt{N}}. \quad (5.30)$$

Taylor's formula in u therefore yields

$$T_E(f) = g_E(1) = g_E(0) + g'_E(0) + O_f(N^{-1/2}). \quad (5.31)$$

It remains to replace $\tilde{T}_E(f)$ by $T_E(f)$. Consider, for example,

$$H_A(u) = \sqrt{N} \nu_u [Q_{12}^- f(Y)].$$

Differentiating with respect to u produces one additional cavity overlap. Using (5.9),

$$|H'_A(u)| \leq \frac{C_f}{\sqrt{N}}.$$

The same argument applies to the B and C classes. Hence

$$\sqrt{N} \nu_0 [Q_{e(E)}^- f(Y)] = \sqrt{N} \nu [Q_{e(E)}^- f(Y)] + O_f(N^{-1/2}). \quad (5.32)$$

At $u = 1$,

$$Q_{e(E)} - Q_{e(E)}^- = \frac{1}{N} \varepsilon_{e(E)},$$

so

$$\sqrt{N} \left| \nu [(Q_{e(E)} - Q_{e(E)}^-) f(X_{12})] \right| \leq \frac{\|f\|_\infty}{\sqrt{N}}.$$

Furthermore,

$$|f(X_{12}) - f(Y)| \leq \frac{2\|f'\|_\infty}{\sqrt{N}},$$

and therefore, by (5.8),

$$\sqrt{N} \nu \left[|Q_{e(E)}^-| |f(X_{12}) - f(Y)| \right] \leq \frac{C_f}{\sqrt{N}}.$$

Combining these estimates,

$$\tilde{T}_E(f) = T_E(f) + O_f(N^{-1/2}), \quad E \in \{A, B, C\}. \quad (5.33)$$

Equations (5.22), (5.29), (5.31), and (5.33) give the approximate Stein system

$$T(f) = \theta m_f + MT(f) + O_f(N^{-1/2}), \quad m_f = \nu[f'(X_{12})]. \quad (5.34)$$

Diagonalization in the three cavity modes. For the test-function correlations define

$$U_f := T_A(f) - 4T_B(f) + 3T_C(f), \quad (5.35)$$

$$V_f := 2T_B(f) - 3T_C(f), \quad (5.36)$$

$$D_f := T_A(f) - 2T_B(f) + T_C(f). \quad (5.37)$$

The same deterministic change of basis as in the cavity closure triangularizes the matrix (5.28). We record the coefficient calculation explicitly.

First,

$$\begin{aligned} \theta_A - 4\theta_B + 3\theta_C &= (1 - q^2) - 4(q - q^2) + 3(r - q^2) \\ &= 1 - 4q + 3r = \kappa, \end{aligned}$$

$$\begin{aligned} 2\theta_B - 3\theta_C &= 2(q - q^2) - 3(r - q^2) \\ &= 2q + q^2 - 3r = \zeta, \end{aligned}$$

$$\begin{aligned} \theta_A - 2\theta_B + \theta_C &= (1 - q^2) - 2(q - q^2) + (r - q^2) \\ &= 1 - 2q + r = a. \end{aligned}$$

Second, direct multiplication of the rows in (5.35)–(5.37) by M gives

$$\begin{aligned} U(MT) &= \beta^2 \kappa U_f, \\ V(MT) &= \beta^2 \zeta U_f + \beta^2 \kappa V_f, \\ D(MT) &= \beta^2 a D_f = \alpha D_f. \end{aligned}$$

Thus (5.34) is equivalent to

$$(1 - \beta^2 \kappa) U_f = \kappa m_f + O_f(N^{-1/2}), \quad (5.38)$$

$$(1 - \beta^2 \kappa) V_f - \beta^2 \zeta U_f = \zeta m_f + O_f(N^{-1/2}), \quad (5.39)$$

$$(1 - \alpha) D_f = a m_f + O_f(N^{-1/2}). \quad (5.40)$$

Using (5.1),

$$U_f = \frac{\kappa}{1 - \beta^2 \kappa} m_f + O_f(N^{-1/2}), \quad (5.41)$$

$$D_f = \frac{a}{1 - \alpha} m_f + O_f(N^{-1/2}). \quad (5.42)$$

Substituting (5.41) into (5.39),

$$\begin{aligned} V_f &= \frac{\zeta m_f + \beta^2 \zeta U_f}{1 - \beta^2 \kappa} + O_f(N^{-1/2}) \\ &= \frac{\zeta}{1 - \beta^2 \kappa} \left(1 + \frac{\beta^2 \kappa}{1 - \beta^2 \kappa} \right) m_f + O_f(N^{-1/2}) \\ &= \frac{\zeta}{(1 - \beta^2 \kappa)^2} m_f + O_f(N^{-1/2}). \end{aligned}$$

The inverse mode transformation is

$$T_A(f) = -V_f - 2U_f + 3D_f. \quad (5.43)$$

Therefore

$$\begin{aligned} \nu[X_{12}f(X_{12})] &= \left(\frac{3a}{1 - \alpha} - \frac{2\kappa}{1 - \beta^2 \kappa} - \frac{\zeta}{(1 - \beta^2 \kappa)^2} \right) \nu[f'(X_{12})] + O_f(N^{-1/2}) \\ &= \sigma^2 \nu[f'(X_{12})] + O_f(N^{-1/2}). \end{aligned} \quad (5.44)$$

This is the required approximate Gaussian integration-by-parts identity.

Notice that the choice $f \equiv 1$ gives

$$\nu[X_{12}] = O(N^{-1/2}), \quad (5.45)$$

so the centering by the infinite-volume parameter q has no residual order-one bias.

Identification and nonnegativity of the variance. For completeness, we check that (5.2) is the limiting second moment. From the cavity equations at $s = 1$, the already proved estimate

$$\nu[|Q_{12}|^3] = o(N^{-1})$$

implies that the cavity remainders are $o(N^{-1})$. Therefore

$$NU \longrightarrow \frac{\kappa}{1 - \beta^2 \kappa}. \quad (5.46)$$

Next,

$$(1 - \beta^2 \kappa)NV - \beta^2 \zeta NU = \zeta + o(1),$$

and (5.46) gives

$$\begin{aligned} NV &\longrightarrow \frac{\zeta + \beta^2 \zeta \kappa / (1 - \beta^2 \kappa)}{1 - \beta^2 \kappa} \\ &= \frac{\zeta}{(1 - \beta^2 \kappa)^2}. \end{aligned} \quad (5.47)$$

Likewise,

$$ND \longrightarrow \frac{a}{1 - \alpha}. \quad (5.48)$$

Since

$$A = -V - 2U + 3D,$$

we conclude that

$$\begin{aligned} N\nu[Q_{12}^2] = NA &\longrightarrow -\frac{\zeta}{(1 - \beta^2 \kappa)^2} - \frac{2\kappa}{1 - \beta^2 \kappa} + \frac{3a}{1 - \alpha} \\ &= \sigma^2. \end{aligned} \quad (5.49)$$

In particular,

$$\sigma^2 \geq 0. \quad (5.50)$$

Characteristic-function ODE. Define

$$C_N(t) := \nu[\cos(tX_{12})], \quad (5.51)$$

$$S_N(t) := \nu[\sin(tX_{12})]. \quad (5.52)$$

For fixed N these functions are differentiable and

$$C'_N(t) = -\nu[X_{12} \sin(tX_{12})], \quad (5.53)$$

$$S'_N(t) = \nu[X_{12} \cos(tX_{12})]. \quad (5.54)$$

Fix $T > 0$. For $|t| \leq T$, the functions

$$f_t(x) = \sin(tx), \quad \tilde{f}_t(x) = \cos(tx)$$

satisfy

$$\|f_t\|_\infty, \|\tilde{f}_t\|_\infty \leq 1, \quad \|f'_t\|_\infty, \|\tilde{f}'_t\|_\infty \leq T, \quad \|f''_t\|_\infty, \|\tilde{f}''_t\|_\infty \leq T^2.$$

Hence the error in (5.44) is uniform for $|t| \leq T$.

Applying (5.44) to f_t gives

$$\nu[X_{12} \sin(tX_{12})] = \sigma^2 t C_N(t) + O_T(N^{-1/2}),$$

and therefore

$$C'_N(t) + \sigma^2 t C_N(t) = O_T(N^{-1/2}) \quad (|t| \leq T). \quad (5.55)$$

Applying (5.44) to $\tilde{f}_t$ gives

$$\nu[X_{12} \cos(tX_{12})] = -\sigma^2 t S_N(t) + O_T(N^{-1/2}),$$

and hence

$$S'_N(t) + \sigma^2 t S_N(t) = O_T(N^{-1/2}) \quad (|t| \leq T). \quad (5.56)$$

Multiply (5.55) by $\exp(\sigma^2 t^2/2)$. Since

$$C_N(0) = 1,$$

we obtain, for $|t| \leq T$,

$$\begin{aligned} \left| e^{\sigma^2 t^2/2} C_N(t) - 1 \right| &\leq \int_0^{|t|} e^{\sigma^2 u^2/2} \frac{C_T}{\sqrt{N}} du \\ &\leq \frac{C_T T}{\sqrt{N}} e^{\sigma^2 T^2/2}. \end{aligned}$$

Therefore

$$\sup_{|t| \leq T} \left| C_N(t) - e^{-\sigma^2 t^2/2} \right| \longrightarrow 0. \quad (5.57)$$

Similarly, since $S_N(0) = 0$,

$$\left| e^{\sigma^2 t^2/2} S_N(t) \right| \leq \int_0^{|t|} e^{\sigma^2 u^2/2} \frac{C_T}{\sqrt{N}} du,$$

so

$$\sup_{|t| \leq T} |S_N(t)| \longrightarrow 0. \quad (5.58)$$

Since $T > 0$ was arbitrary, (5.57) and (5.58) prove (5.3) and (5.4). Equivalently,

$$\nu[e^{itX_{12}}] \longrightarrow e^{-\sigma^2 t^2/2} \quad (t \in \mathbb{R}).$$

The right-hand side is the characteristic function of $\mathcal{N}(0, \sigma^2)$ by (5.50). The continuity theorem for characteristic functions now gives

$$X_{12} = \sqrt{N} (R_{12} - q) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

which completes the proof. $\square$

A Proof of the cavity approximations

Proof of Proposition 3.4. All implicit constants in this proof depend only on $\mathcal{K}$. Write

$$\varepsilon_a := \sigma_N^a, \quad Q_{ab}^- := \frac{1}{N} \sum_{i < N} \sigma_i^a \sigma_i^b - q,$$

so that

$$Q_{ab} = Q_{ab}^- + \frac{1}{N} \varepsilon_a \varepsilon_b. \quad (\text{A.1})$$

By site exchangeability after averaging over the disorder,

$$A_s = \nu_s[Q_{12}(\varepsilon_1\varepsilon_2 - q)], \quad B_s = \nu_s[Q_{12}(\varepsilon_1\varepsilon_3 - q)], \quad C_s = \nu_s[Q_{12}(\varepsilon_3\varepsilon_4 - q)].$$

Use the last-spin cavity interpolation. Write $\sigma = (\rho, \varepsilon)$, with $\rho \in \{-1, 1\}^{N-1}$, and replace the interaction of the last spin with the first $N - 1$ spins by

$$\frac{\beta\sqrt{su}}{\sqrt{N}} \sum_{i < N} \frac{g_{iN} + g_{Ni}}{\sqrt{2}} \rho_i \varepsilon + \beta\sqrt{s(1-u)q} \widehat{z} \varepsilon, \quad 0 \leq u \leq 1,$$

where $\widehat{z}$ is an independent standard Gaussian. Denote the corresponding averaged Gibbs expectation by $\nu_{s,u}$. Then $\nu_{s,1} = \nu_s$. At $u = 0$, the last spin is independent of the first $N - 1$ spins and is subject to the field

$$h + \beta\sqrt{(1-s)q} z_N + \beta\sqrt{sq} \widehat{z} \stackrel{d}{=} h + \beta\sqrt{q} Z.$$

Thus, for distinct replica indices,

$$\mathbb{E}_0[\varepsilon_a \varepsilon_b] = q, \quad \mathbb{E}_0[\varepsilon_a \varepsilon_b \varepsilon_c \varepsilon_d] = r. \quad (\text{A.2})$$

Gaussian integration by parts gives, for every bounded function F of n replicas,

$$\begin{aligned} \frac{d}{du} \nu_{s,u}[F] &= s\beta^2 \left\{ \sum_{1 \leq a < b \leq n} \nu_{s,u}[F \varepsilon_a \varepsilon_b Q_{ab}^-] - n \sum_{a=1}^n \nu_{s,u}[F \varepsilon_a \varepsilon_{n+1} Q_{a,n+1}^-] \right. \\ &\quad \left. + \frac{n(n+1)}{2} \nu_{s,u}[F \varepsilon_{n+1} \varepsilon_{n+2} Q_{n+1,n+2}^-] \right\}. \end{aligned} \quad (\text{A.3})$$

Indeed, the derivative of the covariance of the interpolating last-spin field is $s\beta^2 \varepsilon \varepsilon' Q^-(\rho, \rho')$, while its diagonal value is independent of the configuration.

Set $M_3(u) := \nu_{s,u}[|Q_{12}^-|^3]$. Formula (A.3) and $|Q_{ab}^-| \leq 2$ give

$$|M_3'(u)| \leq C_K M_3(u).$$

Since

$$|Q_{12}^-|^3 \leq 4|Q_{12}|^3 + \frac{4}{N^3},$$

Gronwall's inequality, applied backward from $u = 1$, yields

$$\sup_{0 \leq u \leq 1} \nu_{s,u}[|Q_{12}^-|^3] \leq C_K (N^{-3/2} + \nu_s[|Q_{12}|^3]). \quad (\text{A.4})$$

For

$$P^- = Q_{ab}^- Q_{cd}^-, \quad (ab, cd) \in \{(12, 12), (12, 13), (12, 34)\},$$

the derivative of $\nu_{s,u}[P^-]$ is a finite linear combination of expectations of three cavity overlaps. Hence Hölder's inequality and (A.4) show that

$$|\nu_{s,0}[P^-] - \nu_{s,1}[P^-]| \leq C_K (N^{-3/2} + \nu_s[|Q_{12}|^3]).$$

Moreover, (A.1) gives

$$|P^- - Q_{ab} Q_{cd}| \leq \frac{|Q_{ab}| + |Q_{cd}|}{N} + \frac{1}{N^2}.$$

Using

$$\frac{|x|}{N} \leq \frac{|x|^3}{3} + \frac{2}{3N^{3/2}},$$

we obtain

$$|\nu_{s,0}[P^-] - \nu_s[Q_{ab}Q_{cd}]| \leq C_K(N^{-3/2} + \nu_s[|Q_{12}|^3]). \quad (\text{A.5})$$

Define

$$A_s^- := \nu_{s,0}[(Q_{12}^-)^2], \quad B_s^- := \nu_{s,0}[Q_{12}^-Q_{13}^-], \quad C_s^- := \nu_{s,0}[Q_{12}^-Q_{34}^-],$$

and form U_s^-, V_s^-, D_s^- from A_s^-, B_s^-, C_s^- as in (3.18). It follows from (A.5) that

$$|U_s^- - U_s| + |V_s^- - V_s| + |D_s^- - D_s| \leq C_K(N^{-3/2} + \nu_s[|Q_{12}|^3]). \quad (\text{A.6})$$

Next write

$$\begin{aligned} A_s &= \nu_s[G_A] + \frac{1}{N}\nu_s[E_A], \\ B_s &= \nu_s[G_B] + \frac{1}{N}\nu_s[E_B], \\ C_s &= \nu_s[G_C] + \frac{1}{N}\nu_s[E_C], \end{aligned} \quad (\text{A.7})$$

where

$$\begin{aligned} G_A &:= Q_{12}^-(\varepsilon_1\varepsilon_2 - q), & E_A &:= \varepsilon_1\varepsilon_2(\varepsilon_1\varepsilon_2 - q), \\ G_B &:= Q_{12}^-(\varepsilon_1\varepsilon_3 - q), & E_B &:= \varepsilon_1\varepsilon_2(\varepsilon_1\varepsilon_3 - q), \\ G_C &:= Q_{12}^-(\varepsilon_3\varepsilon_4 - q), & E_C &:= \varepsilon_1\varepsilon_2(\varepsilon_3\varepsilon_4 - q). \end{aligned}$$

Put

$$g_j(u) := \nu_{s,u}[G_j], \quad e_j(u) := \nu_{s,u}[E_j], \quad j \in \{A, B, C\},$$

and take the same linear combinations as in (3.18):

$$g_U := g_A - 4g_B + 3g_C, \quad g_V := 2g_B - 3g_C, \quad g_D := g_A - 2g_B + g_C,$$

with analogous definitions for e_U, e_V, e_D .

Independence at $u = 0$ gives $g_j(0) = 0$. Applying (A.3) twice, Hölder's inequality and (A.4) give

$$g_X(1) = g'_X(0) + C_K(N^{-3/2} + \nu_s[|Q_{12}|^3]), \quad X \in \{U, V, D\}. \quad (\text{A.8})$$

For a replica edge $\{a, b\}$, (A.2) implies

$$\mathbb{E}_0[(\varepsilon_p\varepsilon_q - q)\varepsilon_a\varepsilon_b] = \begin{cases} 1 - q^2, & \{a, b\} = \{p, q\}, \\ q - q^2, & \{a, b\} \text{ shares one endpoint with } \{p, q\}, \\ r - q^2, & \{a, b\} \cap \{p, q\} = \emptyset. \end{cases} \quad (\text{A.9})$$

Substitution into (A.3), followed directly by the three linear combinations above, gives

$$\begin{aligned} g'_U(0) &= s\beta^2\kappa U_s^-, \\ g'_V(0) &= s\beta^2(\zeta U_s^- + \kappa V_s^-), \\ g'_D(0) &= s\alpha D_s^-. \end{aligned} \quad (\text{A.10})$$

The last-spin moments also give

$$\begin{aligned} e_U(0) &= (1 - q^2) - 4(q - q^2) + 3(r - q^2) = \kappa, \\ e_V(0) &= 2(q - q^2) - 3(r - q^2) = \zeta, \\ e_D(0) &= 1 - 2q + r = \frac{\alpha}{\beta^2}. \end{aligned} \tag{A.11}$$

Since E_A, E_B, E_C are uniformly bounded, (A.3) yields

$$|e'_X(u)| \leq C_K \nu_{s,u}[|Q_{12}^-|], \quad X \in \{U, V, D\}.$$

The inequality $\frac{|Q_{12}^-|}{N} \leq \frac{|Q_{12}^-|^3}{3} + \frac{2}{3N^{3/2}}$ and (A.4) therefore imply

$$\frac{1}{N} |e_X(1) - e_X(0)| \leq C_K (N^{-3/2} + \nu_s[|Q_{12}|^3]), \quad X \in \{U, V, D\}. \tag{A.12}$$

From (A.6), (A.7), (A.8), (A.10), (A.11), and (A.12), it follows that

$$\begin{aligned} U_s &= s\beta^2 \kappa U_s + \frac{\kappa}{N} + C_K (N^{-3/2} + \nu_s[|Q_{12}|^3]), \\ V_s &= s\beta^2 (\zeta U_s + \kappa V_s) + \frac{\zeta}{N} + C_K (N^{-3/2} + \nu_s[|Q_{12}|^3]), \\ D_s &= s\alpha D_s + \frac{\alpha}{\beta^2 N} + C_K (N^{-3/2} + \nu_s[|Q_{12}|^3]). \end{aligned}$$

These yield (3.20)–(3.23). □

B Finite-step proof of the specialized Guerra–Talagrand bound

B.1 A finite-step differentiation formula

The following lemma isolates the only hierarchical Gaussian calculation needed in the proof.

Lemma B.1. *Let $\mathcal{S}$ be finite, let μ be a nonzero finite measure on $\mathcal{S}$, and fix*

$$0 = m_0 \leq m_1 \leq \dots \leq m_M \leq m_{M+1} = 1.$$

Let $\{G_{k,t}(\xi) : \xi \in \mathcal{S}, t \in [0, 1]\}_{0 \leq k \leq M}$ be independent centered Gaussian processes with almost surely C^1 sample paths in $t \in (0, 1)$, and let $K_{k,t}$ denote their covariance kernels. Assume that $K_{k,t}$ is bounded. Put

$$\mathcal{G}_t(\xi) := \sum_{k=0}^M G_{k,t}(\xi), \quad C_t^{(j)}(\xi, \eta) := \sum_{k=0}^j K_{k,t}(\xi, \eta).$$

For a deterministic function $V : \mathcal{S} \rightarrow \mathbb{R}$, define

$$X_M(t) = \log \sum_{\xi \in \mathcal{S}} \mu(\xi) e^{\mathcal{G}_t(\xi) + V(\xi)}$$

and, recursively for $j = M, \dots, 1$,

$$X_{j-1}(t) := \begin{cases} \frac{1}{m_j} \log \mathbb{E}_j e^{m_j X_j(t)}, & m_j > 0, \\ \mathbb{E}_j X_j(t), & m_j = 0, \end{cases} \tag{B.1}$$

where $\mathbb{E}_j$ integrates only the field $G_{j,t}$. For $0 \leq j \leq M$, set

$$\mathcal{I}_j A := \frac{\mathbb{E}_j[Ae^{m_j X_j(t)}]}{\mathbb{E}_j e^{m_j X_j(t)}},$$

and let

$$p_t(\xi) := \frac{\mu(\xi)e^{\mathcal{G}_t(\xi)+V(\xi)}}{\sum_{\eta \in \mathcal{S}} \mu(\eta)e^{\mathcal{G}_t(\eta)+V(\eta)}}.$$

For deterministic functions F and $0 \leq j \leq M$, define

$$\begin{aligned} \mathbb{E}\langle F(\xi) \rangle_t &:= \sum_{\xi \in \mathcal{S}} F(\xi) \mathcal{I}_0 \cdots \mathcal{I}_M p_t(\xi), \\ \mathbb{E}\langle F(\xi, \eta) \rangle_{j,t} &:= \sum_{\xi, \eta \in \mathcal{S}} F(\xi, \eta) \mathcal{I}_0 \cdots \mathcal{I}_j \left[(\mathcal{I}_{j+1} \cdots \mathcal{I}_M p_t(\xi)) (\mathcal{I}_{j+1} \cdots \mathcal{I}_M p_t(\eta)) \right], \end{aligned}$$

where an empty operator product is the identity. Then, for $0 < t < 1$,

$$\frac{d}{dt} \mathbb{E}_0 X_0(t) = \frac{1}{2} \mathbb{E} \left\langle \dot{C}_t^{(M)}(\xi, \xi) \right\rangle_t - \frac{1}{2} \sum_{j=0}^M (m_{j+1} - m_j) \mathbb{E} \left\langle \dot{C}_t^{(j)}(\xi, \eta) \right\rangle_{j,t} \quad (\text{B.2})$$

where $\dot{C}_t^{(j)}(\xi, \eta) := \frac{\partial}{\partial t} C_t^{(j)}(\xi, \eta)$.

Proof. Throughout the proof, for fixed t , we regard $X_j(t)$ as a smooth function of the Gaussian coordinates

$$\{G_{k,t}(\xi) : 0 \leq k \leq j, \xi \in \mathcal{S}\}$$

that have not yet been integrated out; the fields at levels $j+1, \dots, M$ have already been averaged out by the recursion. Thus $\partial_x X_{j-1} = \mathcal{I}_j(\partial_x X_j)$ whenever x is a Gaussian coordinate not integrated by $\mathbb{E}_j$ or the time coordinate t . Indeed, if $Z_j = \mathbb{E}_j e^{m_j X_j}$ and $m_j > 0$, then

$$\partial_x X_{j-1} = \frac{1}{m_j} \frac{\partial_x Z_j}{Z_j} = \frac{\mathbb{E}_j[e^{m_j X_j} \partial_x X_j]}{Z_j} = \mathcal{I}_j(\partial_x X_j). \quad (\text{B.3})$$

The same identity for $m_j = 0$ follows by differentiating $X_{j-1} = \mathbb{E}_j X_j$. From (B.3) it follows that

$$\partial_{G_{k,t}(\xi)} X_k = \mathcal{I}_{k+1} \cdots \mathcal{I}_M p_t(\xi). \quad (\text{B.4})$$

For any differentiable A , the quotient rule gives the more general identity

$$\partial_y \mathcal{I}_j A = \mathcal{I}_j(\partial_y A) + m_j \mathcal{I}_j(A \partial_y X_j) - m_j \mathcal{I}_j(A) \mathcal{I}_j(\partial_y X_j). \quad (\text{B.5})$$

Consequently,

$$\partial_{xy} X_{j-1} = \mathcal{I}_j(\partial_{xy} X_j) + m_j \{ \mathcal{I}_j(\partial_x X_j \partial_y X_j) - \mathcal{I}_j(\partial_x X_j) \mathcal{I}_j(\partial_y X_j) \}. \quad (\text{B.6})$$

Again, this includes $m_j = 0$, since the last two terms then vanish.

By (B.3), we have

$$\begin{aligned} \frac{d}{dt} \mathbb{E} X_0 &= \mathcal{I}_0 \cdots \mathcal{I}_M \frac{d}{dt} X_M(t) \\ &= \sum_{k=0}^M \mathcal{I}_0 \cdots \mathcal{I}_M \left(\sum_{\xi \in \mathcal{S}} p_t(\xi) \dot{G}_{k,t}(\xi) \right) \\ &= \sum_{k=0}^M \sum_{\xi \in \mathcal{S}} \mathcal{I}_0 \cdots \mathcal{I}_k \left(\dot{G}_{k,t}(\xi) (\mathcal{I}_{k+1} \cdots \mathcal{I}_M (p_t(\xi))) \right). \end{aligned} \quad (\text{B.7})$$

When $m_k > 0$, by Gaussian integration by parts (see [19, Vol. 1, Sec. A.4]), i.e. for centered jointly Gaussians X, Y_i ,

$$\mathbb{E}[Xf(Y_1 \dots Y_d)] = \sum_{i=1}^d \text{Cov}(X, Y_i) \mathbb{E}[\partial_i f(Y)],$$

where $\text{Cov}(\dot{G}_{k,t}(\xi), G_{k,t}(\eta)) + \text{Cov}(\dot{G}_{k,t}(\eta), G_{k,t}(\xi)) = \dot{K}_{k,t}(\xi, \eta)$, by (B.4) we have

$$\begin{aligned} \sum_{\xi \in \mathcal{S}} \mathcal{I}_k \left(\dot{G}_{k,t}(\xi) (\mathcal{I}_{k+1} \dots \mathcal{I}_M p_t(\xi)) \right) &= \sum_{\xi \in \mathcal{S}} Z_k^{-1} \mathbb{E}_k \left[e^{m_k X_k} \dot{G}_{k,t}(\xi) (\mathcal{I}_{k+1} \dots \mathcal{I}_M p_t(\xi)) \right] \\ &= \frac{1}{2} \sum_{\xi, \eta \in \mathcal{S}} \dot{K}_{k,t}(\xi, \eta) (Z_k)^{-1} \mathbb{E}_k \left[\partial_{G_{k,t}(\eta)} (e^{m_k X_k} (\mathcal{I}_{k+1} \dots \mathcal{I}_M p_t(\xi))) \right] \\ &= \frac{1}{2} \sum_{\xi, \eta \in \mathcal{S}} \dot{K}_{k,t}(\xi, \eta) (m_k Z_k)^{-1} \mathbb{E}_k \left[\partial_{G_{k,t}(\eta)} \partial_{G_{k,t}(\xi)} e^{m_k X_k} \right]. \end{aligned}$$

Let $H_{j,t}(\xi) = \sum_{r=0}^j G_{r,t}(\xi)$. For fixed k , write

$$u_k^\xi = \partial_{H_{k,t}(\xi)} X_k, \quad h_k^{\xi\eta} = \partial_{H_{k,t}(\xi)} \partial_{H_{k,t}(\eta)} X_k.$$

The contribution obtained by differentiating only the covariance of $G_{k,t}$ is

$$(m_k Z_k)^{-1} \mathbb{E}_k [\partial_{G_k(\xi)} \partial_{G_k(\eta)} e^{m_k X_k}] = \mathcal{I}_k \left(h_k^{\xi\eta} + m_k u_k^\xi u_k^\eta \right).$$

For $m_k = 0$, the same formula follows directly from (B.7), with the second summand equal to zero. At $k = 0$ it also holds because $\mathcal{I}_0 = \mathbb{E}_0$ and $m_0 = 0$. Hence, letting

$$\mathcal{P}_k = \mathcal{I}_0 \mathcal{I}_1 \dots \mathcal{I}_k, \quad Q_k^{\xi\eta} = h_k^{\xi\eta} + m_k u_k^\xi u_k^\eta,$$

by (B.7) we have

$$\frac{d}{dt} \mathbb{E} X_0 = \frac{1}{2} \sum_{k=0}^M \sum_{\xi, \eta \in \mathcal{S}} \dot{K}_{k,t}(\xi, \eta) \mathcal{P}_k Q_k^{\xi\eta}. \quad (\text{B.8})$$

It remains to compute $\mathcal{P}_k Q_k^{\xi\eta}$. At the terminal level, direct differentiation gives

$$u_M^\xi = p_t(\xi), \quad h_M^{\xi\eta} = \mathbf{1}_{\{\xi=\eta\}} p_t(\xi) - p_t(\xi) p_t(\eta).$$

Since $m_{M+1} = 1$, it follows that

$$Q_M^{\xi\eta} = \mathbf{1}_{\{\xi=\eta\}} p_t(\xi) - (m_{M+1} - m_M) u_M^\xi u_M^\eta. \quad (\text{B.9})$$

Note that (B.4) implies $\partial_{H_{j-1,t}(\xi)} X_j = \partial_{H_{j,t}(\xi)} X_j$. For $j = M, \dots, 1$, equations (B.3) and (B.6) yield

$$u_{j-1}^\xi = \mathcal{I}_j u_j^\xi, \quad h_{j-1}^{\xi\eta} = \mathcal{I}_j h_j^{\xi\eta} + m_j \left(\mathcal{I}_j (u_j^\xi u_j^\eta) - u_{j-1}^\xi u_{j-1}^\eta \right),$$

and hence

$$Q_{j-1}^{\xi\eta} = h_{j-1}^{\xi\eta} + m_{j-1} u_{j-1}^\xi u_{j-1}^\eta = \mathcal{I}_j Q_j^{\xi\eta} - (m_j - m_{j-1}) u_{j-1}^\xi u_{j-1}^\eta.$$

By applying $\mathcal{P}_{j-1}$ to both sides of this equality, we have

$$\mathcal{P}_{j-1} Q_{j-1}^{\xi\eta} = \mathcal{P}_j Q_j^{\xi\eta} - (m_j - m_{j-1}) \mathcal{P}_{j-1} (u_{j-1}^\xi u_{j-1}^\eta).$$

Iterating from $j = M$ down to $j = k + 1$ and using (B.9) gives

$$\mathcal{P}_k Q_k^{\xi\eta} = \mathcal{P}_M (\mathbf{1}_{\{\xi=\eta\}} p_t(\xi)) - \sum_{j=k}^M (m_{j+1} - m_j) \mathcal{P}_j(u_j^\xi u_j^\eta). \quad (\text{B.10})$$

Note that similarly to (B.4) it holds that $u_j^\xi = \mathcal{I}_{j+1} \cdots \mathcal{I}_M p_t(\xi)$. Substituting (B.10) into (B.8) therefore give

$$\begin{aligned} \frac{d}{dt} \mathbb{E} X_0 &= \frac{1}{2} \mathbb{E} \left\langle \sum_{k=0}^M \dot{K}_{k,t}(\xi, \xi) \right\rangle_t - \frac{1}{2} \sum_{k=0}^M \sum_{j=k}^M (m_{j+1} - m_j) \mathbb{E} \left\langle \dot{K}_{k,t}(\xi, \eta) \right\rangle_{j,t} \\ &= \frac{1}{2} \mathbb{E} \left\langle \dot{C}_t^{(M)}(\xi, \xi) \right\rangle_t - \frac{1}{2} \sum_{j=0}^M (m_{j+1} - m_j) \mathbb{E} \left\langle \dot{C}_t^{(j)}(\xi, \eta) \right\rangle_{j,t}, \end{aligned}$$

where we have used $\sum_{k=0}^M \sum_{j=k}^M \dot{K}_{k,t}(\xi, \eta) = \sum_{j=0}^M \sum_{k=0}^j \dot{K}_{k,t}(\xi, \eta) = \sum_{j=0}^M \dot{C}_t^{(j)}(\xi, \eta)$. $\square$

B.2 Application to the two-replica path

Fix $N \geq 1$, $\beta > 0$, $h \in \mathbb{R}$, $s \in [0, 1]$, and $q \in (0, 1)$. We use the same notation as in Section 2.2. Recall that

$$\xi_s(r) = \beta^2(1-s)qr + \frac{s\beta^2}{2}r^2.$$

Let $H_{N,s}^{\text{cen}}$ be a centered Gaussian field on Σ_N normalized by

$$\mathbb{E} H_{N,s}^{\text{cen}}(\sigma) H_{N,s}^{\text{cen}}(\tau) = N \xi_s(R(\sigma, \tau)). \quad (\text{B.11})$$

Set

$$H_{N,s}(\sigma) = H_{N,s}^{\text{cen}}(\sigma) + h \sum_{i=1}^N \sigma_i, \quad \mathcal{R}_N = \{-1, -1 + 2/N, \dots, 1\}.$$

For $v \in \mathcal{R}_N$, let

$$Z_{N,s}^{(2)}(v) = \sum_{\substack{\sigma^1, \sigma^2 \in \Sigma_N \\ R(\sigma^1, \sigma^2) = v}} e^{H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2)}. \quad (\text{B.12})$$

We prove (2.38) as follows.

Lemma B.2. *For every $N \geq 1$ and $v \in \mathcal{R}_N$,*

$$\frac{1}{N} \mathbb{E} \log Z_{N,s}^{(2)}(v) \leq \inf_{\lambda \in \mathbb{R}} \mathfrak{P}_s(\lambda, v). \quad (\text{B.13})$$

Proof. In view of (2.25) and (2.33), we are able to choose independent Gaussian vectors $z_{i,j} \in \mathbb{R}^2$ and independent Gaussian variables y_j with

$$\mathbb{E} z_{i,j} z_{i,j}^\top = B_j - B_{j-1}, \quad \mathbb{E} y_j^2 = d_j - d_{j-1}. \quad (\text{B.14})$$

Let $Y_{i,0}$ be independent copies of $Y_0 = h + \beta\sqrt{(1-s)q} Z$. Since

$$B_0 = \beta^2(1-s)q \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

the vector $(Y_{i,0} - h, Y_{i,0} - h)$ has the covariance B_0 . Write

$$\mathcal{S}_N(v) = \{(\sigma^1, \sigma^2) \in \Sigma_N^2 : R(\sigma^1, \sigma^2) = v\}.$$

For $\sigma = (\sigma^1, \sigma^2) \in \mathcal{S}_N(v)$, introduce the independent Gaussian random variables $\{G_{j,t}(\sigma)\}_{j=0}^M$ with parameter $t \in [0, 1]$ as

$$\begin{aligned} G_{0,t}(\sigma) &:= \sqrt{t}(H_{N,s}^{\text{cen}}(\sigma^1) + H_{N,s}^{\text{cen}}(\sigma^2)) + \sqrt{1-t} \sum_{i=1}^N \sum_{a=1}^2 (Y_{i,0} - h) \sigma_i^a, \\ G_{j,t}(\sigma) &:= \sqrt{1-t} \sum_{i=1}^N \sum_{a=1}^2 z_{i,j}^a \sigma_i^a + \sqrt{tN} y_j, \quad 1 \leq j \leq M, \end{aligned}$$

and the deterministic term

$$V(\sigma) := h \sum_{i=1}^N (\sigma_i^1 + \sigma_i^2) + \lambda \sum_{i=1}^N \sigma_i^1 \sigma_i^2 - \lambda N v.$$

Let

$$X_M(t) := \log \sum_{\sigma \in \mathcal{S}_N(v)} \exp \left(\sum_{j=0}^M G_{j,t}(\sigma) + V(\sigma) \right),$$

define $\{X_j(t)\}_{j=0}^M$ as in Lemma B.1, and set

$$\varphi_N(t) := \frac{1}{N} \mathbb{E} X_0(t).$$

We first compute $\varphi_N(t)$ at the endpoints $t = 0, 1$. Note that

$$\begin{aligned} X_M(0) &= \log \sum_{\sigma \in \mathcal{S}_N(v)} \exp \left(\sum_{i=1}^N Y_{i,0} (\sigma_i^1 + \sigma_i^2) + \sum_{j=1}^M \sum_{i=1}^N (z_{i,j}^1 \sigma_i^1 + z_{i,j}^2 \sigma_i^2) + \lambda \sum_{i=1}^N \sigma_i^1 \sigma_i^2 - \lambda N v \right) \\ &\leq \log \sum_{\sigma \in \Sigma_N^2} \exp \left(\sum_{i=1}^N Y_{i,0} (\sigma_i^1 + \sigma_i^2) + \sum_{j=1}^M \sum_{i=1}^N (z_{i,j}^1 \sigma_i^1 + z_{i,j}^2 \sigma_i^2) + \lambda \sum_{i=1}^N \sigma_i^1 \sigma_i^2 - \lambda N v \right) \\ &= \log \prod_{i=1}^N \sum_{\varepsilon_1, \varepsilon_2 = \pm 1} \exp \left(\left(\varepsilon_1 \left(Y_{0,i} + \sum_{j=1}^M z_{i,j}^1 \right) + \varepsilon_2 \left(Y_{0,i} + \sum_{j=1}^M z_{i,j}^2 \right) + \lambda \varepsilon_1 \varepsilon_2 \right) \right) - \lambda N v \\ &= 2N \log 2 + \sum_{i=1}^N f_\lambda \left(Y_{0,i} + \sum_{j=1}^M z_{i,j}^1, Y_{0,i} + \sum_{j=1}^M z_{i,j}^2 \right) - \lambda N v. \end{aligned}$$

Since G_j in Section 2.2 has the same law as $z_{i,j}$ (see (B.14)), by the monotonicity of the map $\mathcal{T}_{m,C}$ defined in (2.26), similarly to (2.31) we obtain

$$X_0(0) \leq 2N \log 2 + \sum_{i=1}^N U_s^{\lambda,v}(Y_{0,i}, Y_{0,i}) - \lambda N v.$$

This inequality and the independence of the variables $Y_{0,i}$ yield

$$\varphi_N(0) \leq 2 \log 2 + \mathbb{E} U_s^{\lambda,v}(Y_0, Y_0) - \lambda v. \quad (\text{B.15})$$

We consider the case $t = 1$. An explicit calculation implies

$$\begin{aligned} X_M(1) &= \log \sum_{\sigma \in \mathcal{S}_N(v)} \exp \left(H_{N,s}(\sigma^1) + H_{N,s}(\sigma^2) + \sqrt{N} \sum_{j=1}^M y_j \right) \\ &= \log Z_{N,s}^{(2)}(v) + \sqrt{N} \sum_{j=1}^M y_j, \end{aligned}$$

where $Z_{N,s}^{(2)}(v)$ is defined in (B.12). Since for each $x \in \mathbb{R}$ it holds that

$$\begin{cases} \frac{1}{m_j} \log \mathbb{E} e^{m_j(x + \sqrt{N} y_j)} = x + \frac{N m_j}{2} (d_j - d_{j-1}), & m_j > 0, \\ \mathbb{E}(x + \sqrt{N} y_j) = x, & m_j = 0, \end{cases}$$

by (B.1) we have

$$\begin{aligned} \varphi_N(1) &= \frac{1}{N} \mathbb{E} \log Z_{N,s}^{(2)}(v) + \frac{1}{2} \sum_{j=1}^M m_j (d_j - d_{j-1}) \\ &= \frac{1}{N} \mathbb{E} \log Z_{N,s}^{(2)}(v) + \mathcal{L}_s(v), \end{aligned} \tag{B.16}$$

where $\mathcal{L}_s(v)$ is defined in (2.34).

We calculate the derivative of φ_N . Denote the covariance of $\sum_{i=0}^j G_{i,t}(\sigma)$ and $\sum_{i=0}^j G_{i,t}(\tau)$ by $C_t^{(j)}(\sigma, \tau)$. Since (B.11) and $\mathbb{E}(Y_{0,i} - h)^2 = \beta^2(1-s)q$ imply

$$\mathbb{E}[G_{0,t}(\sigma)G_{0,t}(\tau)] = tN \sum_{k,\ell=1}^2 \xi_s(R(\sigma^k, \tau^\ell)) + (1-t)\beta^2(1-s)qN \sum_{k,\ell=1}^2 R(\sigma^k, \tau^\ell)$$

and (B.14) implies

$$\begin{aligned} \mathbb{E}[G_{j,t}(\sigma)G_{j,t}(\tau)] &= (1-t) \sum_{i=1}^N \sum_{k,\ell=1}^2 \sigma_i^k \tau_i^\ell \mathbb{E}[z_{i,j}^k z_{i,j}^\ell] + tN \mathbb{E} y_j^2 \\ &= (1-t) \sum_{i=1}^N \sigma_i^\top (B_j - B_{j-1}) \tau_i + tN(d_j - d_{j-1}), \end{aligned}$$

by (2.23) and (2.32) we have

$$\begin{aligned} C_t^{(j)}(\sigma, \tau) &= tN \sum_{k,\ell=1}^2 \xi_s(R(\sigma^k, \tau^\ell)) + (1-t)\beta^2(1-s)qN \sum_{k,\ell=1}^2 R(\sigma^k, \tau^\ell) \\ &\quad + (1-t) \sum_{i=1}^N \sigma_i^\top (B_j - B_0) \tau_i + tN(d_j - d_0) \\ &= tN \sum_{k,\ell=1}^2 \xi_s(R(\sigma^k, \tau^\ell)) + (1-t)N \sum_{k,\ell=1}^2 \xi'_s(Q_{j,k\ell}^v) R(\sigma^k, \tau^\ell) + \frac{s\beta^2 tN}{2} \sum_{k,\ell=1}^2 (Q_{j,k\ell}^v)^2 \\ &= N \sum_{k,\ell=1}^2 \xi'_s(Q_{j,k\ell}^v) R(\sigma^k, \tau^\ell) + t \left(\frac{s\beta^2 N}{2} \sum_{k,\ell=1}^2 (R(\sigma^k, \tau^\ell) - Q_{j,k\ell}^v)^2 \right), \end{aligned}$$

due to $\xi_s(x) - \xi_s(y) - \xi'_s(y)(x - y) = \frac{s\beta^2}{2}(x - y)^2$. Since for $\sigma \in \mathcal{S}_N(v)$

$$(R(\sigma^k, \sigma^\ell))_{k,\ell=1}^2 = \begin{pmatrix} 1 & v \\ v & 1 \end{pmatrix} = Q_M^v,$$

it follows that

$$\dot{C}_t^{(M)}(\sigma, \sigma) = \frac{s\beta^2 N}{2} \sum_{k,\ell=1}^2 (R(\sigma^k, \sigma^\ell) - Q_{M,k\ell}^v)^2 = 0.$$

Hence, Lemma B.1 now yields

$$\varphi'_N(t) = -\frac{s\beta^2}{4} \sum_{j=0}^M (m_{j+1} - m_j) \mathbb{E} \left\langle \sum_{k,\ell=1}^2 (R(\sigma^k, \tau^\ell) - Q_{j,k\ell}^v)^2 \right\rangle_{j,t} \leq 0. \quad (\text{B.17})$$

Combining (B.15), (B.16), and (B.17), and taking the infimum over λ proves (B.13). $\square$

C Price's theorem

Theorem C.1 (Price's theorem [16]). *For each $t \in \mathbb{R}$, let $X(t)$ be an n -dimensional centered Gaussian vector with covariance matrix $C(t) = (C_{ij}(t))_{i,j=1}^n$. Assume that $C_{ij}(t)$ is continuously differentiable for all $i, j = 1, 2, \dots, n$. Then, for any $f \in C_b^2(\mathbb{R}^n)$, it holds that*

$$\frac{d}{dt} \mathbb{E} f(X_t) = \frac{1}{2} \sum_{i,j=1}^n \left(\frac{d}{dt} C_{i,j}(t) \right) \mathbb{E} \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(X(t)) \right].$$

Proof. It suffices to prove the claim for compactly supported $f \in C_b^2(\mathbb{R}^n)$. Let $\mathcal{F}g$ be the Fourier transform of g , i.e.

$$(\mathcal{F}g)(\xi) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} g(x) e^{-\sqrt{-1}x \cdot \xi} dx, \quad \xi \in \mathbb{R}^n,$$

and let $C^\varepsilon(t) := \varepsilon I + C(t)$ where $\varepsilon > 0$ and I is the unit matrix. Denote an n -dimensional centered Gaussian vector with a covariance matrix $C^\varepsilon(t)$ by $X^\varepsilon(t)$ for $t \in \mathbb{R}$. Then, by the Plancherel equality we have

$$\begin{aligned} \mathbb{E} f(X^\varepsilon(t)) &= \int_{\mathbb{R}^n} f(x) \cdot \frac{1}{\sqrt{(2\pi)^n \det C^\varepsilon(t)}} \exp \left(-\frac{1}{2} \langle x, C^\varepsilon(t)^{-1} x \rangle \right) dx \\ &= \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} (\mathcal{F}f)(\xi) \exp \left(-\frac{1}{2} \langle \xi, C^\varepsilon(t) \xi \rangle \right) d\xi, \end{aligned}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product on $\mathbb{R}^n$ and $\det C^\varepsilon(t)$ denotes the determinant of $C^\varepsilon(t)$. Since $X^\varepsilon(t)$ converges in distribution to $X(t)$ and $C^\varepsilon(t)$ converges componentwise to $C(t)$ as $\varepsilon \downarrow 0$, differentiating with respect to t and then passing to the limit $\varepsilon \downarrow 0$ yields

$$\frac{d}{dt} \mathbb{E} f(X(t)) = -\frac{1}{2(2\pi)^{n/2}} \sum_{i,j=1}^n \left(\frac{d}{dt} C_{i,j}(t) \right) \int_{\mathbb{R}^n} (\mathcal{F}f)(\xi) \xi_i \xi_j \exp \left(-\frac{1}{2} \langle \xi, C(t) \xi \rangle \right) d\xi,$$

where $\xi = (\xi_1, \xi_2, \dots, \xi_n)$. Applying the fact

$$\xi_i \xi_j (\mathcal{F}f)(\xi) = -\mathcal{F} \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right) (\xi),$$

to this equality, we obtain

$$\frac{d}{dt}\mathbb{E}f(X(t)) = \frac{1}{2(2\pi)^{n/2}} \sum_{i,j=1}^n \left(\frac{d}{dt} C_{i,j}(t) \right) \int_{\mathbb{R}^n} \mathcal{F} \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right) (\xi) \exp \left(-\frac{1}{2} \langle \xi, C(t) \xi \rangle \right) d\xi. \quad (\text{C.1})$$

Similarly to above, it holds that

$$\begin{aligned} & \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \mathcal{F} \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right) (\xi) \exp \left(-\frac{1}{2} \langle \xi, C(t) \xi \rangle \right) d\xi \\ &= \lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}^n} \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right) (x) \cdot \frac{1}{\sqrt{(2\pi)^n \det C^\varepsilon(t)}} \exp \left(-\frac{1}{2} \langle x, C^\varepsilon(t)^{-1} x \rangle \right) dx \\ &= \mathbb{E} \left[\frac{\partial^2 f}{\partial x_i \partial x_j} (X(t)) \right]. \end{aligned}$$

This equality and (C.1) yield the assertion. $\square$

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Statement on the Use of Artificial Intelligence

Shortly after posting [7], inspired by the arguments in [18], the authors realized that the flatness of the Guerra-Talagrand one-step RSB bound near the replica-symmetric point is sufficient to obtain a preliminary weak concentration estimate for the overlap, as explained in Subsection 1.3. This weak concentration can be upgraded, using Talagrand’s cavity method, to the optimal N^{-1} bound. Thus, the main remaining task was to establish the required flatness of the one-step RSB bound.

To investigate this flatness, we asked ChatGPT 5.6 Sol to construct a proof, with reference to Lopatto’s paper [10]. After several iterations, ChatGPT produced an argument that led to the proof presented in this paper. The authors subsequently checked and revised the argument in detail and take full responsibility for its correctness. We also used Codex to formalize the proof in Lean 4 [8].

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